Worksheet 1.1 Atomic Theory	

NAME: **CLASS:**

INTRODUCTION

One way to represent a particular atom is with a **nuclide** symbol:

$${}^{A}_{Z}X$$

where A = mass number, Z = atomic number and X = element symbol
Often this notation is used to represent only a nucleus. For the questions below, assume that the atoms represented also contain electrons.

No.	Question	Answer
1	${}^{252}_{98}\mathrm{Cf}$ is an isotope used to treat cervical cancer. How many of each of the following does one atom of ${}^{252}_{98}\mathrm{Cf}$ contain? **a** protons **b** neutrons	
2	One isotope of nickel used in the detection of explosives has a mass number of 63. Write a nuclide symbol to represent this isotope.	
3	How many protons, neutrons and electrons does each of the following ions contain? **a** ${}^{31}_{15}\mathrm{P}^{3-}$ **b** ${}^{235}_{92}\mathrm{U}^{2+}$	
4	What is the nuclide symbol for carbon-12?	
5	How are the two isotopes of gallium, ${}^{69}_{31}\mathrm{Ga}$ and ${}^{71}_{31}\mathrm{Ga}$, similar in: **a** atomic structure? **b** chemical properties?	

Radioactive atoms

The nuclei of some atoms, particularly those composed of a large number of protons and neutrons, are often unstable. These radioactive atoms spontaneously break down or disintegrate by emitting at least one of three different types of radiation – alpha (α), beta (β) and gamma (γ) radiation – to become more stable. Alpha particles are helium nuclei and consist of two protons and two neutrons. Beta particles are electrons that come from the breaking down of neutrons. Gamma rays are electromagnetic energy.

Atoms with atomic numbers greater than 82 are radioactive, with most of these decaying by emitting alpha radiation. For example, americium-241 is the radioactive isotope used in smoke detectors and it decays according to the equation: $^{241}_{95}Am \rightarrow ^{4}_{2}He + ^{237}_{93}Np$.

Light atoms that have too many neutrons decay by emitting beta particles. In this decay process, a neutron in the nucleus transforms into a proton and an electron. The proton stays in the nucleus and the electron is ejected energetically. Carbon-14, which makes less than 1% of all naturally occurring carbon atoms, undergoes beta decay to form the nitrogen isotope of $^{14}_{7}N$: $^{14}_{6}C \rightarrow ^{14}_{7}N + e^{-}$

Carbon-14 is used in the carbon-dating process to date fossils and rocks.

No.	Question	Answer
6	Explain why the decay of a radioactive isotope by either alpha-decay or beta-decay produces an isotope of a different element.	
7	The americium-241 isotope does not occur naturally on Earth. It is formed in nuclear reactors when plutonium-241 undergoes beta decay. Describe, in terms of protons and neutrons, what happens during this nuclear reaction.	
8	Thorium-229, which is used to prolong the life of fluorescent light tubes, undergoes alpha-decay. Give the equation to represent this decay.	

Worksheet 1.2 Electron configurations	

NAME: **CLASS:**

INTRODUCTION

In the quantum mechanical model, electrons are considered to exist in energy levels or shells. These energy levels are denoted by the numbers 1, 2, 3, 4, … From quantum mechanics it has been calculated that the maximum number of electrons that a shell can accommodate is $2n^2$, where n is the shell number.

Examples of electron arrangements or configurations in terms of shells are:

oxygen (8): 2, 6 and potassium (19): 2, 8, 8, 1.

No.	Question	Answer
1	Which shell contains electrons of lower energy, the first shell or the second shell?	
2	Which shell contains electrons that are on average closer to the nucleus, the first shell or the third shell?	
3	What is the maximum number of electrons that can occupy the second shell?	
4	Write the electron configurations for neutral atoms of: **a** sulfur (16) **b** nitrogen (7) **c** calcium (20).	
5	Write the electron configurations for the: **a** chloride ion (Cl^-) **b** aluminium ion (Al^{3+}) **c** sulfide ion (S^{2-}) **d** potassium ion (K^+).	
6	Germanium is in group 14 of the periodic table. How does this relate to its electron configuration?	

Worksheet 1.2 Electron configurations	

IONISATION ENERGY

The ionisation energy of an atom or ion is defined as the amount of energy required to remove the most loosely bound electron from the atom or ion in the gaseous state.

The energy required to remove the first electron from an atom is called the first ionisation energy. The second ionisation energy is the energy required to remove the second electron, and so on. These are called successive ionisation energies. The successive ionisation energies for magnesium are shown in Table 1.1.

Ionisation	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th	12th
Ionisation energy (MJ mol^{-1})	0.74	1.46	7.74	10.55	13.64	18.00	21.71	25.66	31.65	35.47	170.00	189.38

No.	Question	Answer
7	Why is energy required to remove electrons from an atom?	
8	What is the electron configuration of magnesium?	
9	Suggest why more energy is required to remove the second electron compared to removing the first electron.	
10	For magnesium, suggest why much more energy is required to remove the third electron than is required to remove the second electron.	
11	Why would you expect the 11th ionisation energy to be much larger than the 10th ionisation energy?	
12	Explain which of the substances, sodium, aluminium or silicon, you expect to have the following first seven successive ionisation energies: 584, 1 823, 2 751, 11 584, 14 837, 18 384, 23 302 kJ mol^{-1}.	

Worksheet 1.3 Famous scientists	

NAME: **CLASS:**

INTRODUCTION

The history of chemistry is incomplete without acknowledging the ingenuity and brilliance of the scientists involved. No single country monopolised the discoveries, with progress coming from a range of European countries. Choose some of the following questions to guide your research into the lives of these famous scientists.

No.	Question	Answer
1	How did Michael Faraday obtain his first chemistry job? What discoveries are attributed to Faraday?	
2	In what ways was Henry Cavendish eccentric? What discoveries are attributed to him?	
3	Einstein made contributions to chemistry as well as to physics. What contributions did Einstein make to chemistry?	
4	How did Ernest Rutherford come to go to England? What is transmutation and why is Rutherford relevant to any discussion of transmutation?	
5	What country was Amedeo Avogadro from? What did he conclude about many of the common gases such as hydrogen and oxygen?	
6	Jons Berzelius made many contributions to chemistry. What was he most famous for?	

7	What is ‘Daltonism’ and where did the term come from?	
8	By what name is Maria Sklodowska better known? What did she actually discover?	
9	Who was the ‘Geber’ and what did he discover?	
10	When was the mass spectrometer invented? Who invented it, and how was it helpful?	
11	Who was Julius Rosenberg, and how did he die?	
12	How did Niels Bohr explain the observation that many elements emit light when electricity is passed through them?	

Worksheet 1.4 Developing a periodic table	

NAME: **CLASS:**

INTRODUCTION

During the nineteenth century, as the amount of chemical information quickly increased, some scientists began to classify this information in useful ways. Dmitri Mendeleev was one of these scientists. He used these card-playing techniques to help him order the elements into a periodic table. He wrote the names and properties of the elements on cards and tried many different arrangements of them. For this worksheet, you will need to cut out about 30 small pieces of cardboard on which to write the names of elements.

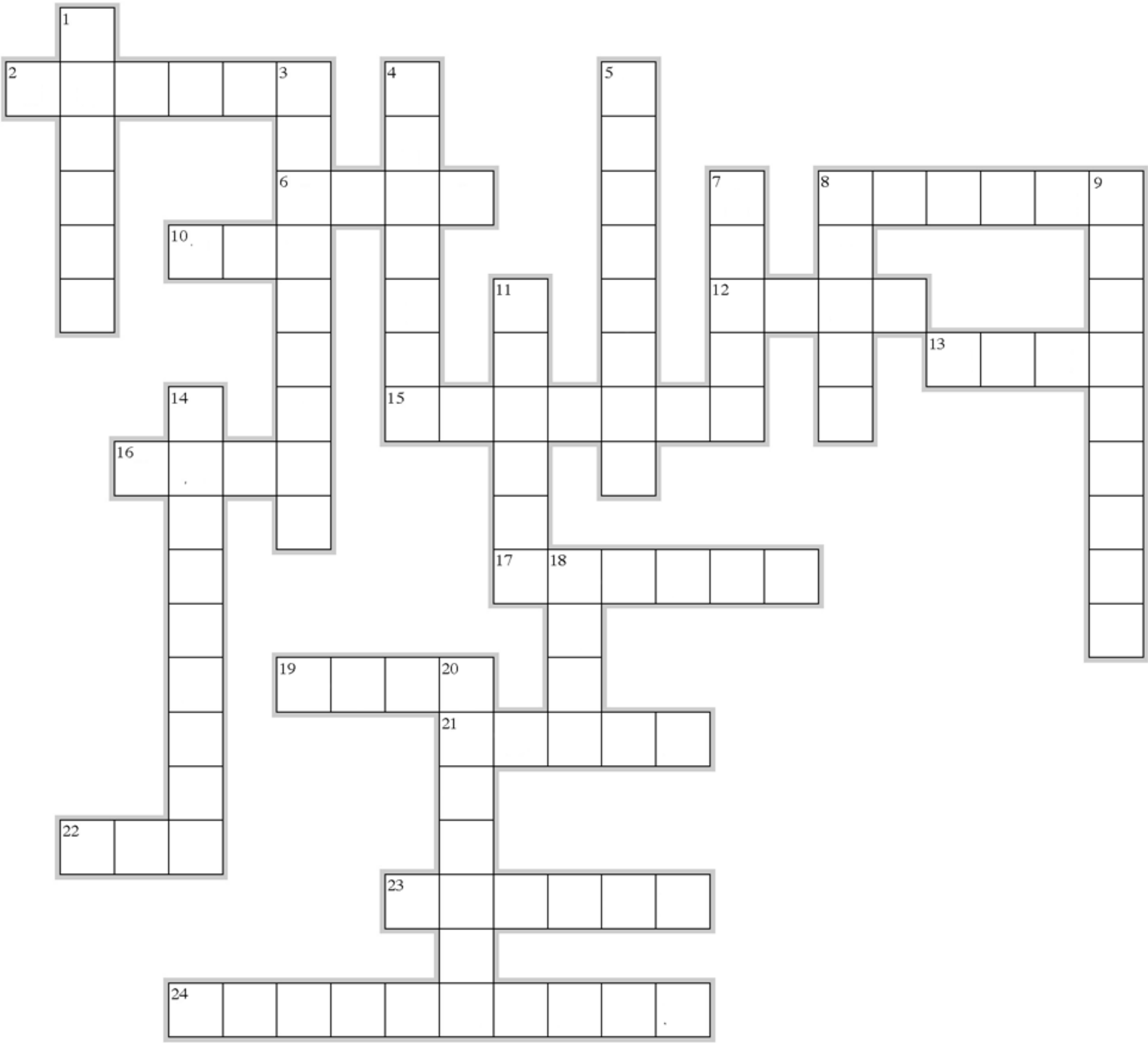
No	Question	Answer
1	Write each of these symbols on separate cards: C, S, Cu, Zn, Au. Can you think of a connection between these elements?	
2	Add the following symbols to further cards: Cl, H, C, Be, B, F, Li, Na, Mg, K, Ca, N, O, Al, P, Br. These are examples of elements commonly available in the early 1800s. Can you place any of these elements into groups with some connecting theme? What basis did you use for your groups?	
3	Add to each card the rounded off relative atomic mass of each of these elements. Now lay the cards out in order of their mass. Is this the way the elements are ordered in the modern periodic table?	
4	John Newlands decided that the properties of the elements repeated themselves every eighth element. Do this with your elements. Line up the hydrogen and then the next seven elements. Put sodium under hydrogen and keep adding the next elements until you complete the second row. Comment on the consistency of the columns you have formed.	

5	What did Mendeleev do to solve the problem described in question **4**?	
6	Now line up the elements so that the columns match the modern periodic table. Name some missing elements.	
7	Why does sodium behave like lithium and potassium?	
8	Argon was the first group 18 element isolated (by Sir William Ramsay). Make a card and place it in its rightful place. Why did Ramsay think there was more than one inert gas?	
9	How does the position of an element in a vertical group in the modern periodic table relate to the element's electron configuration?	
10	The representation of the modern periodic table we use is only one possibility. Use the Internet to research alternative forms of the periodic table. Sketch one of these alternate forms.	

Worksheet 2.1

Metals – history, uses and properties

NAME: **CLASS:**



ACROSS

2. The metal ion present in table salt.
6. Which metal completes the quote ‘all that glitters is not …?’
8. The age dating from 3000 BC is named after this metal.
10. Copper and this metal combine to form bronze.
12. A soft, dense metal formerly used in plumbing.
13. A reactive metal often used in torch batteries.
15. A liquid metal associated with mad hatters.
16. Tin-plated iron is used to make these useful containers.
17. Combines with copper to produce Australian ‘silver’ coins.
19. This form of aluminium is used in cooking.
21. This type of bond exists in the compound formed when a metal reacts with a non-metal.
22. An economically useful mineral.
23. One of the precious metals.
24. They sought to change lead to gold.

DOWN

1. One of the metals that is not silver in colour.
3. Iron and some other metals exhibit this property.
4. An important metal ion for healthy bones and teeth.
5. The ninth most abundant element in the Earth’s crust.
7. A mixture of metals.
8. An alloy of copper and zinc.
9. The conductivity of metals is due to the movement of these particles.
11. This element is added to iron to produce steel.
14. A property shared by all metals
18. The metal ion used to transport oxygen around your body.
20. The metal with the smallest atomic number.

Worksheet 2.2	
Modifying metals	

Worksheet 2.2
Modifying metals

NAME: **CLASS:**

INTRODUCTION

Metals in their pure forms have many uses. For example, tungsten is used in light globe filaments, aluminium for cans and foil, and mercury in thermometers. More often than not, however, different metals are mixed together to produce an **alloy**, which is more useful than the pure metals. An alloy usually contains at least two different metals, and sometimes one or more non-metals. Alloys include:

Common name	Components (% by mass)		Uses
Rose gold (18 carat)	Gold Copper Silver	75% 22.25% 2.75%	Jewellery
Stainless steel	Iron Nickel Chromium Carbon	74% 8% 18% less than 1%	Sinks, cutlery
Brass	Copper Zinc	65% 35%	Musical instruments, sculptures
Solder	Tin Lead	67% 33%	Joining metal parts together

No.	Question	Answer
1	Ordinary steel is composed of iron, with less than 1% carbon added for strength. Why would ordinary steel be unsuitable for cutlery and sinks?	
2	Food tins are usually made of steel coated with tin. What properties must tin have?	
3	Pure gold is no longer used to make jewellery. What benefits do alloys offer in jewellery making?	
4	What are two properties that solder must have?	
5	What are three properties that make brass suitable for use in musical instruments?	

Worksheet 2.2

Modifying metals

Many metals undergo heat treatments during manufacturing. Sometimes this is done to make the metal softer, sometimes to make it harder. The heat treatment changes the metallic crystal structure. For example, pure aluminium is too weak and ductile for such purposes as aeroplane parts. However, when alloyed with 4% copper and appropriately heat-treated, it becomes very strong. This alloy is called duralumin. It is very light and strong. Many aluminium alloys are used in aeroplanes.

No.	Question	Answer
6	Explain why alloying aluminium would produce a less ductile product.	
7	Titanium alloys are even lighter and stronger than aluminium alloys. Why then are most aircraft made from aluminium alloys, and not titanium alloys?	
8	What are two advantages in a commercial aircraft being as light as possible?	
9	Some parts of aeroplanes need to be made from more flexible alloys. What is one part that would need some degree of flexibility?	
10	During the production of duralumin, the aluminium alloy is quenched. Consult books or the internet to find out what 'quenching' is. Explain why it makes the metal harder.	

Worksheet 3.1	
Ionic solids	

NAME: **CLASS:**

INTRODUCTION

This worksheet looks at the formation and bonding of ionic compounds and their properties.

No.	Question	Answer
1	Reactions between which of the following pairs of elements would be expected to form ionic compounds? **a** Carbon and oxygen **b** Tin and chlorine **c** Sodium and fluorine **d** Zinc and copper	
2	After considering the electron configuration of the following metals, determine the number of electrons each atom will lose, and the formula of the ion they will form when they react with a non-metal.	
3	After considering the electron configuration of the following non-metals, determine the number of electrons each atom will gain, and the formula of the ion they will form when they react with a metal.	
4	Give the formula of the ionic compound formed between the following pairs of ions: **a** Cu^{2+} and F^{-} **b** Al^{3+} and O^{2-}	
5	Element A, from group 13, reacts with element B, from group 17, to form a compound. What is the formula of this compound?	
6	An ionic compound has the formula A_3B_2. Elements A and B could be found in which groups of the periodic table?	

Question 2 table:

Metal	Number of electrons lost	Formula of ion formed
Calcium		
Potassium		

Question 3 table:

Non-metal	Number of electrons gained	Formula of ion formed
Chlorine		
Nitrogen		

7	Draw electron dot diagrams to represent the transfer of electrons and the number of atoms required to produce each of the following ionic compounds. **a** Sodium oxide from the reaction of sodium with oxygen atoms. **b** Magnesium sulfide from the reaction of magnesium with sulfur.	
8	Assuming that the ions are arranged in the same way, which of the following ionic compounds would have the higher melting point: KF or CaO? Explain your reasoning.	
9	Sodium chloride is an excellent conductor of electricity when dissolved in water, but in the solid state does not conduct at all. Explain this observation in terms of the ionic bonding model.	
10	Sodium is a soft malleable substance, yet sodium chloride is a hard brittle substance. Explain why NaCl is brittle but Na is malleable.	

Worksheet 3.2 Electrovalencies	

NAME: **CLASS:**

INTRODUCTION

Knowledge of the electrovalencies (charges) of ions is essential for any student of chemistry. This worksheet provides practice in determining electrovalencies, and in writing chemical formulas.

No.	Question	Answer
1	Write the formula of each of the following ions: **a** Sulfate ion **b** Calcium ion **c** Nitrate ion **d** Nitrite ion **e** Fluoride ion **f** Lead(II) ion	
2	Write the name of each of the following ions: **a** Cr^{2+} **b** NH_4^+ **c** Sn^{4+} **d** CN^- **e** Mn^{2+} **f** $H_2PO_4^-$	
3	Write the names of the following ions: **a** CO_3^{2-} **b** SO_4^{2-} **c** NO_3^- **d** PO_4^{3-} **e** MnO_4^- What do all of these names have in common?	
4	List eight anions that finish in the letters '-ide'. What do all of these ions have in common?	

5	Metal X forms an ionic oxide with the formula X_2O_3. In which group of the periodic table is X most likely to be found?	
6	Write the chemical formula for each of the following compounds: **a** Potassium hydroxide **b** Lithium dichromate **c** Strontium iodide **d** Copper(II) phosphate **e** Magnesium hydrogensulfate **f** Aluminium oxalate	
7	Manganese can form a range of compounds, including $Mn(NO_3)_2$, Mn_2O_3 and MnO_2. List the electrovalencies shown by manganese in these compounds.	
8	Write the names of the following compounds. **a** BaO **b** $Cu(NO_3)_2$ **c** $Sn(OH)_2$ **d** $NaHCO_3$ **e** $Fe(H_2PO_4)_2$ **f** Sr_3N_2	
9	Determine the formula of the positive ion for each of the following compounds: **a** Radium oxide (RaO) **b** Tellurium chloride ($TeCl_6$) **c** Uranium sulfate ($U_2(SO_4)_3$) **d** Gold tetrafluoride ion (AuF_4^-) **e** Lanthanum nitride (LaN)	
10	Give the formula and name of the negative ion for each of the following compounds: **a** Calcium selenite ($CaSeO_3$) **b** Potassium cyanate ($KCNO$) **c** Aluminium bromate ($Al(BrO_3)_3$)	

Worksheet 4.1 Electron dot diagrams of molecules	

Worksheet 4.1
Electron dot diagrams of molecules

NAME: **CLASS:**

INTRODUCTION

Covalent bonding occurs between non-metal atoms and involves the sharing of one or more electrons between these atoms to (generally) ensure that each atom shares eight outer-shell electrons (octet rule). Multiple bonds may be formed to ensure that the octet rule is met. Electron dot diagrams can be used to represent molecules.

No.	Question	Answer
1	Complete the table below by placing a tick in the relevant column to indicate the type or types of bonding exhibited by each substance:	

Substance	Ionic	Covalent	Metallic
Copper			
Oxygen (O_2)			
Ammonia (NH_3)			
Calcium sulfide			
Potassium carbonate			

No.	Question	Answer
2	Which of the following list of substances can be accurately described as being composed of molecules? **a** Helium **b** Glucose ($C_6H_{12}O_6$) **c** Brass **d** Nitrogen monoxide **e** Aluminium chloride	
3	Draw electron dot diagrams for the following molecules, and so determine the number of bonding and non-bonding electron pairs in each.	

Molecule	F_2	CO_2	H_2S
Electron dot diagram			
Bonding pairs			
Non-bonding pairs			

4	Explain why, when carbon combines with hydrogen, the molecules formed have a formula of CH_4 and not CH_3 or CH_5. Include an electron dot diagram in your answer.	
5	Draw electron dot diagrams for the following:	
6	Choose a substance from the list below to match its description. • Ethyne (C_2H_2) • Trichloromethane ($CHCl_3$) • Sulfur dioxide (SO_2) • Phosphorus trichloride (PCl_3) **a** Contains three single bonds **b** Nine lone pairs of electrons **c** Contains a double bond **d** Contains a triple bond	
7	Consider the molecules O_2 and N_2. Which of these two molecules has the: **a** longer bond length? **b** stronger bond?	
8	**a** Draw an electron dot diagram for $CaCO_3$. **b** Describe the bonding in this compound.	

H_2O	H_2O_2	OH^-	NO_3^-

Worksheet 4.2
Polythene

NAME: **CLASS:**

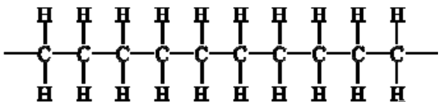
INTRODUCTION

Polythene

Polythene, or poly(ethene), is used extensively as the plastic in plastic bags for fruit and vegetables, garbage bin liners, squeeze bottles, cling wrap and many other items.

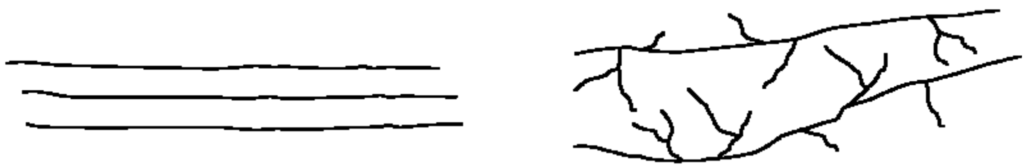
The very large molecules making up polythene make it a little different in structure from most other covalent molecular compounds. Yet it still behaves like a covalent molecular compound. It melts at temperatures between 120 and 140^{o}C, it is flexible and does not conduct electricity.

As with all polymers, the very large molecules of polythene are made up of repeating units. These repeating units are $–CH_2–$ as shown in the following line structure:



A polythene molecule can contain up to 100 000 repeating units.

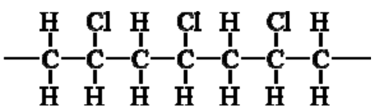
Polythene is prepared by reacting the monomer ethene, $CH_2 = CH_2$, in the presence of a catalyst. Monomers are the small molecules used to make polymers. Two different kinds of polythene, a low density polymer and a high density polymer, can be made depending on the conditions used to form them. Low density polythene has a lower melting point, is more flexible and more transparent, but is not as strong as the high density polythene. The low density form is the one used for the plastic bags, but the high density form is used for water pipes and electrical cable insulation. The differences between these two forms of the same polymer are due to the variation in the structures of their chains. In high density polythene, the chains are very linear, but in low density polythene, the chains are branched, as shown below, where a line is used to represent the chains of CH_2.



linear chains **branched chains**

Branched chains cannot fit as closely together as the linear chains.

All of the plastic materials that are commonly used, such as polystyrene, bakelite, Teflon and PVC, are polymers similar to polythene. They have the same long chains of carbon atoms joined one to the other, but they have some different groups joined to this chain. For example PVC has chlorine atoms covalently bonded to some of the carbon atoms in the long chains as shown below



Worksheet 4.2 Polythene	

No.	Question	Answer
1	Propose a reason why the form of polythene with the branched chains is called low density polythene.	
2	**a** Explain why a plastic bag melts easily if touched with a hot object. **b** Suggest why low density polythene has a lower melting point than high density polythene?	
3	Suggest why high density polythene is less flexible than low density polythene.	
4	Despite there being many electrons in a molecule of polythene, it does not conduct electricity. Explain why.	
5	Draw an electron dot diagram and a line structure of ethene. (In a line structure, a single line represents a pair of electrons.)	
6	Teflon, the non-stick coating sometimes used on saucepans, is prepared from the monomer tetrafluorethene, C_2F_4. **a** Draw an electron dot diagram and a line structure of tetrafluoroethene. **b** Draw a line structure of a small portion of the polymer Teflon.	
7	Draw a line structure and an electron dot diagram of the monomer used to produce the PVC molecules shown on page 1.	

Worksheet 5.1 Covalent network lattices	

NAME: **CLASS:**

INTRODUCTION

This worksheet looks at the covalent network lattices of diamond, graphite, silicon, silicon carbide and silicon dioxide, their properties and structure.

No.	Question	Answer
1	**a** Using lines to represent the bonds between the atoms, draw a small section of the diamond lattice. **b** What type of bond is between the atoms?	
2	Name an element and a compound that have a network lattice structure that is similar to diamond. What properties do these substances share with diamond?	
3	Draw a small section of the graphite layer lattice to demonstrate how it differs from diamond.	
4	Graphite and diamond are allotropic forms of carbon. Explain why graphite conducts electricity whereas diamond does not.	
5	Diamond is the hardest natural substance known, with the maximum rating of 10 on the Mohs scale of hardness. Why is diamond so hard?	

6	State two practical uses of diamond that rely on its extreme hardness.	
7	The oxides of the first two elements in group 14 exhibit dramatically different properties. Carbon dioxide changes state from a solid to a gas (it sublimes) at a temperature of –78°C, whereas silicon dioxide melts at 1700°C. Explain these very different melting points in terms of the structures of the two oxides.	
8	Silicon and phosphorus are non-metal elements adjacent to one another on the periodic table. Silicon has a melting point of 1410°C, compared to just 44°C for phosphorus. Explain this substantial difference.	
9	Graphite is a soft and slippery substance and, as such, can be used as a 'dry' lubricant in racing car engines. Explain these properties of graphite in terms of its structure and bonding.	
10	Nanotubes are a recently developed example of another allotropic form of carbon. In what way is the structure of nanotubes similar to that of: **a** diamond? **b** graphite?	

Worksheet 5.2

A bonding summary

NAME: **CLASS:**

1	Complete the table below by placing a tick in the relevant column to indicate the type of bonding exhibited by each substance.

Substance	Ionic	Covalent	Metallic
Aluminium oxide			
Hydrogen sulfide			
Zinc hydroxide			
Brass			
Graphite			
Tungsten			

2 Classify the bonding present in each of the following substances as ionic, metallic or covalent.

a A hard but malleable substance with a high melting temperature that conducts electricity well in either a solid or molten state.

b A gaseous substance at room temperature that cannot conduct electricity, even when cooled to a liquid.

c A hard but brittle substance with a very high melting point. Dissolves readily in water to produce a solution that conducts electricity.

3 You are provided with a sample of a white solid substance, a magnifying glass, some water, a Bunsen burner, tripod and crucible, and electrical conductivity testing equipment. Explain how you would use some or all of this equipment to determine whether the substance is ionic, metallic or covalent molecular.

4 Complete the following summary table, using the notes below as a guide.

a Use the words: ‘atoms’, ‘ions’, ‘electrons’, ‘cations’ or ‘molecules’.
b Include a simple sketch of each lattice type.
c Use ‘hard’ or ‘soft’, and include ‘brittleness’ or ‘malleability’.
d Use ‘high’, ‘low’, ‘generally high’, ‘generally low’ or ‘very high’.
e Use ‘yes’, ‘no’ or ‘not applicable’.

		Ionic	Metallic	Covalent molecular	Covalent network
a	Particles present				
b	Structure				
c	Hardness (at room temperature)				
d	Melting point				
e	Conductivity:				
	Solid				
	Liquid				
	Aqueous solution				

5 Draw and annotate sketches to show:

a ionic solids are brittle.	**b** metals are malleable.	**c** the change in conductivity when an ionic solid melts.

Worksheet 6.1

Calculation of relative masses

NAME: **CLASS:**

INTRODUCTION

The determination of relative atomic, molecular and formula masses is important, as it allows for calculations involving the mole and, consequently, stoichiometry.

No.	Question	Answer
1	Complete the table below:	

Term	Definition	Symbol	Unit
Relative atomic mass of an element			
	The amount of substance containing the same number of particles as there are atoms in 12 g of carbon-12.		
			g mol^{-1}
Avogadro's constant			

No.	Question	Answer
2	Explain why the relative atomic mass of carbon is 12.01, even though carbon (mass of exactly 12) is used as the standard on which relative mass is determined.	
3	The element sulfur has three isotopes: ^{32}S, ^{33}S and ^{34}S. If their relative atomic masses and percentage abundances are 31.97 (95.0%), 32.97 (0.77%) and 33.97 (4.23%) respectively, estimate what value you would expect the relative atomic mass of the element sulfur to be.	

4	Calculate the relative molecular mass of each of the following compounds: **a** Hydrogen cyanide (HCN) **b** Phosphoric acid (H_3PO_4) **c** Urea ($(NH_2)_2CO$)	
5	Calculate the relative formula mass of each of the following compounds: **a** Aluminium iodide (AlI_3) **b** Calcium carbonate ($CaCO_3$) **c** Ammonium sulfate ($(NH_4)_2SO_4$)	
6	An oxide of nitrogen has a relative molecular mass of 46. What is the formula of this oxide?	
7	Tin exists as +2 and +4 ions in its compounds. Determine the relative formula masses of the two possible tin chlorides.	
8	The relative atomic mass of hydrogen-1 is 1.0 and the relative atomic mass of fluorine is 19.0. **a** How much heavier is a fluorine atom than a hydrogen-1 atom? **b** How much heavier will 100 fluorine atoms be than 100 hydrogen-1 atoms? **c** How much heavier will 6×10^{23} fluorine atoms be than 6×10^{23} hydrogen-1 atoms? **d** If the mass of 1 mole of hydrogen-1 atoms is 1 g, what is the mass of 1 mole of fluorine atoms?	
9	What is the molar mass of **a** zinc? **b** zinc chloride?	

Worksheet 6.2	
Mole calculations	

NAME: **CLASS:**

INTRODUCTION

The mole is one of the most important concepts in chemistry as it allows for the quantitative calculation of amounts of substances that may take part in chemical reactions. Various formulas are used to calculate the amount of substance (in mol):

$n = \frac{m}{M}$, where n = amount in mol, m = mass and M = molar mass (g mol^{-1})

$n = \frac{N}{N_A}$ where N = number of particles, N_A = Avogadro's constant (6.02×10^{23} mol^{-1})

No.	Question	Answer
1	Determine the mass of one mole of each of the following: **a** PbO_2 **b** $CuCl_2$ **c** H_2SO_4	
2	Calculate the amount of substance (in mol) for each of the following: **a** 250 g of barium sulfate ($BaSO_4$) **b** 500 kg of iron(III) oxide (Fe_2O_3) **c** 3.388×10^{-6} g of benzene (C_6H_6) **d** 30.0 tonnes of alumina (Al_2O_3)	
3	Determine the number of **a** electrons in 1.00 mol of electrons **b** atoms in 12.2 mol of zirconium **c** atoms in 120 g of argon **d** molecules in 7.20×10^{-8} g of ozone (O_3).	

4	Calculate the mass of each of the following: **a** 40.0 mol of octane (C_8H_{18}) **b** 2.80×10^{-5} mol of aspartame ($C_{14}H_{18}N_2O_5$) **c** 3.079 mol of calcium arsenate ($Ca_3(AsO_4)_2$)	
5	Determine the amount (in mol) of hydrogen atoms in each of the following: **a** 2.6 mol of hydrogen sulfide (H_2S) **b** $8.1 \text{ x } 10^{-3}$ mol of hexanol ($C_6H_{13}OH$) **c** 35.0 g of beryllium hydride (BeH_2) **d** 0.977 kg of aluminium hydrogenphosphate ($Al_2(HPO_4)_3$)	
6	A compound widely used as a pigment in paints due to its brilliant white colour has the formula XO_2. If 0.693 mole of this substance has a mass of 55.37 g, determine the identity of element X.	
7	An important component of the fast-acting and very effective adhesive, Superglue, is methyl-2-cyanoacrylate, a compound with the chemical formula of $C_5H_5NO_2$. **a** Determine the molar mass of this compound. **b** What mass of methyl-2-cyanoacrylate will be present in 8.22×10^{-4} mol of the compound?	

Worksheet 7.1

The kinetic molecular theory

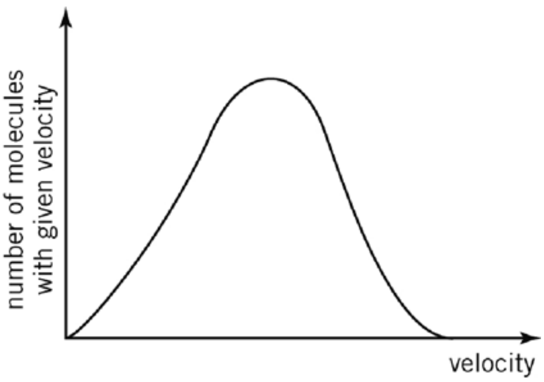
NAME: **CLASS:**

INTRODUCTION

The kinetic molecular theory (KMT) is a model used to explain the behaviour of gases. The essential ideas of this theory are:

- gaseous particles are in constant random motion, colliding with each other and the walls of the container
- the volume of each gas particle is negligible compared to the total gas volume
- forces between gas particles are negligible
- collisions are perfectly elastic; energy may be transferred between particles, but the average kinetic energy of the particles does not change
- average kinetic energy of the particles is proportional to the absolute temperature of the gas.

No.	Question	Answer
1	A sample of carbon dioxide gas is at room temperature and pressure. The sample is then compressed to a smaller volume, while keeping the temperature constant. How will this change the: **a** pressure of the sample? **b** average kinetic energy of the molecules? **c** average speed of the molecules?	
2	A 50 mL container is filled with nitrogen and sealed. The temperature is then raised by 20°C. Explain why the pressure increases when this change in temperature is made.	
3	A short tube with a stopcock connects two flasks of equal volume. One flask contains oxygen at a pressure of 2 atm; the other is evacuated. What will happen when the stopcock is opened?	
4	A syringe with a free-moving plunger contains nitrogen. Explain why the plunger will move in when the syringe is cooled.	

5	A student has two 1.0 L flasks. Flask A contains methane (CH_4); flask B contains oxygen (O_2). Both flasks are at the same temperature and pressure, and so must contain an equal number of molecules. Which of these flasks: **a** contains the greater mass of gas? **b** has molecules with a higher average kinetic energy? **c** has the greater density?	
6	The graph below shows the distribution of molecular velocities in a gas sample at 20°C. On the same graph, sketch the expected curve for the same gas sample at 40°C.  	
7	Some materials, like the rubber that a balloon is made of, have tiny holes through which gases can escape. If two identical balloons are filled with identical volumes of different gases (helium and nitrogen), which balloon would be more deflated after 24 hours? Explain your choice.	
8	The KMT assumes that there are no forces between gas particles. Under what temperature and pressure conditions would forces between particles become significant?	
9	When a helium balloon is released from ground level, explain why its volume increases as it rises in the atmosphere.	

Worksheet 8.1 Water – a glossary	

NAME: **CLASS:**

INTRODUCTION

Chapters 7 and 8 introduce many new terms and ideas about liquids, and in particular, water. To help consolidate your understanding of these terms, complete the following glossary entries. For each entry you should use an annotated diagram and/or chemical equations to show the meaning of the term or descriptive phrase provided. You are *not to write* definitions; only diagrams and equations are required—be creative!

No.	Term/description	Diagram/equation(s)
1	A solution of sodium chloride has a higher boiling point than pure water.	
2	Distillation may be used to separate liquids with different boiling points.	
3	Pure water does not conduct electricity, neither does a mixture of water and silver chloride, but a solution of ammonium chloride does.	
4	Molecules of hydrogen chloride (HCl) ionise when placed in water.	

5	At the same temperature, methylated spirits (ethanol) has a higher vapour pressure than water.	
6	The evaporation of water is utilised as a cooling mechanism by the human body.	
7	The boiling point of water is reduced at high altitudes due to reduced air pressure.	
8	Oil and water are immiscible liquids.	
9	Whilst boiling, the temperature of the boiling water remains constant.	
10	When copper carbonate is added to some water, the liquid remains colourless. However, when copper sulfate is added to water, the liquid becomes a blue colour.	

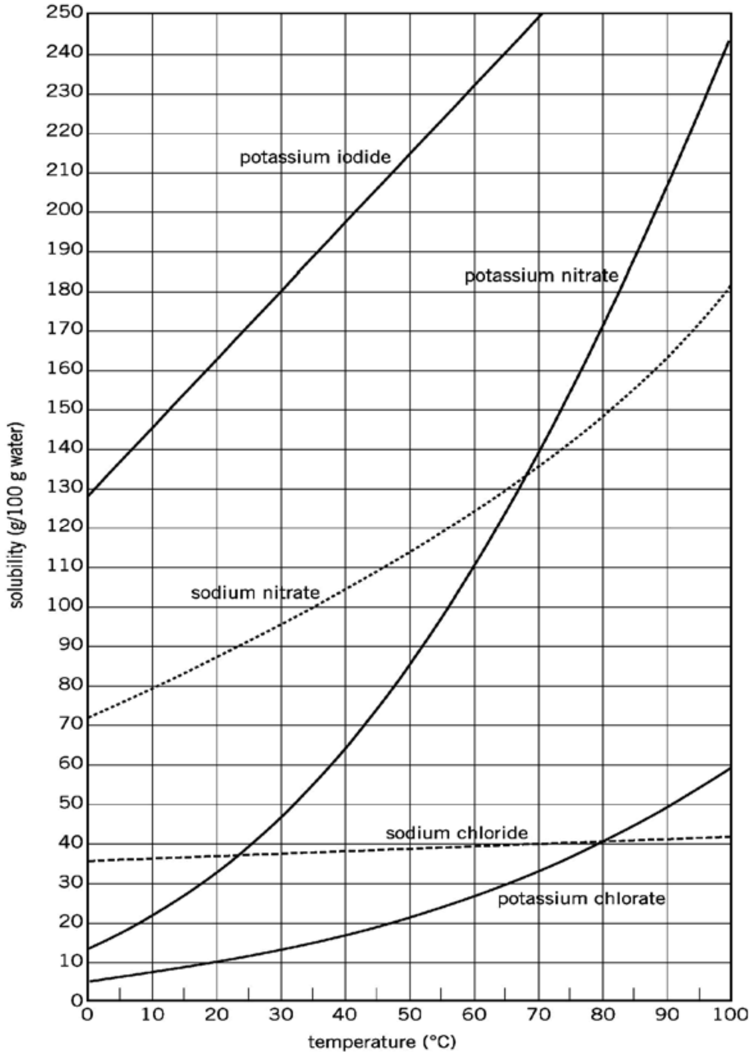
Worksheet 8.2

Solubility curves

NAME: **CLASS:**

INTRODUCTION

The graph below shows the solubility curves for a number of ionic compounds. Use this graph when answering the questions that follow.



No.	Question	Answer
1	What is the solubility of sodium chloride at 75°C?	
2	Which substance is the least soluble at 0°C?	

3	Which substance is the most soluble at 70°C?	
4	Which substance experiences the greatest change in solubility between 0°C and 30°C?	
5	If a solution has 44 g of potassium nitrate dissolved in 100 g of water at 30°C, is it a saturated, unsaturated or supersaturated solution?	
6	300 g of potassium iodide is added to 100 g of water at 40°C. What mass of solid remains undissolved?	
7	What mass of water would have to be added to the solution in question **6** in order to dissolve the remaining solid, assuming the temperature does not change?	
8	A saturated solution is made by dissolving potassium nitrate in 50 g of water at 80°C. If this solution were to be cooled to 20°C, what mass of solid would crystallise out?	
9	What mass of water needs to be added to 100 g of potassium iodide at 50°C to make a saturated solution?	
10	9.0 g of potassium chlorate is added to 60 g of water. At what minimum temperature would all the solid dissolve in the water?	

Worksheet 8.3 Concentration calculations	

NAME: **CLASS:**

INTRODUCTION

The concentration of a solution may be expressed using a variety of units. These units include:

- **% m/m**, which means the mass in gram of solute in 100 g of solution
- **% m/V**, which means the mass in gram of solute in 100 mL of solution
- **% V/V**, which means the volume in mL of solute in 100 mL of solution
- **ppm**, which means the mass in gram of solute in 10^6 g of solution (equivalent to mg L^{-1} for dilute solutions)
- **concentration in mol L^{-1}**, which means the amount in mole of solute in 1.0 L of solution.
- **concentration in g L^{-1}**, which means the mass in gram of solute in 1.0 L of solution.

.

No.	Question	Answer
1	A bottle of cabernet sauvignon wine is labelled as 14% v/v ethanol. What volume of ethanol would be in a 700 mL bottle of this wine?	
2	What is the concentration, in mol L^{-1}, of a solution in which 40.0 g of sucrose ($C_{12}H_{22}O_{11}$) is dissolved in 300 mL of the solution?	
3	What volume of water must be added to 1.00 L of a solution containing 70.2 g of NaCl to produce a solution of 0.670 mol L^{-1} NaCl?	
4	What volume of 5.00% m/V cloudy ammonia cleaning solution is needed to make 250 mL of a 1.50% m/V solution?	

5	30.0 mL of a sodium carbonate solution is made up to a volume of 300 mL with distilled water. The resultant solution has a sodium carbonate concentration of 0.108 mol L^{-1}. What mass of sodium carbonate was present in the original solution?	
6	50 mL of a 2.0% m/V glucose solution is mixed with 50 mL of a 6.0% m/V glucose solution. The solution is then made up to 300 mL with distilled water. What is the concentration (% m/V) of the final solution?	
7	A sample of water from Toxic Lake is found to contain 600 ppm mercury. What mass of mercury is in 5.00 Mg of lake water?	
8	Seawater contains 0.0097 mol L^{-1} K^+ ions. What is this concentration in ppm? (Assume 1 mL of sea water has a mass of 1 g)	
9	A solution of insecticide is made by dissolving 3.0 g of compound Z in distilled water and making the solution up to 2.0 L. After being sprayed on the plants, 2.0 mL of this solution remains in the container. 5.0 L of water is added to flush the container out (assume the volumes are additive). A stray fly then lands on the lip of the container and drinks 1.0 μL of this dilute solution. What mass of compound Z did the fly ingest?	

Worksheet 9.1
Balancing chemical equations

NAME: **CLASS:**

INTRODUCTION

The law of conservation of mass states that 'matter can be neither created nor destroyed, it can only be changed from one form to another'. For chemical equations, this means that there must be the same number of each type of atom on both sides of a chemical equation. For ionic equations, the charge on either side of the arrow must be the same.

Balance the following equations by adding coefficients to the chemical formulas where needed. Add the missing states to each equation (assume the reactions are carried out).

No.	
1	$SO_4^{2-}(aq) + \quad Ag^+(aq) \rightarrow \quad Ag_2SO_4(s)$
2	$Ni(NO_3)_2(s) \rightarrow \quad NiO(s) + \quad NO_2(\ \ ) + \quad O_2(\ \ )$
3	$P(s) + O_2(\ \ ) \rightarrow P_4O_{10}(s)$
4	$CH_3COOH(aq) + \quad K_2CO_3(s) \rightarrow \quad CH_3COO^-(aq) + \quad K^+(aq) + \quad CO_2(\ \ ) + \quad H_2O(l)$
5	$Al(\ \ ) + \quad HI(g) \rightarrow \quad AlI_3(s) + \quad H_2(\ \ )$
6	$C_5H_{12}(g) + \quad O_2(\ \ ) \rightarrow \quad CO_2(\ \ ) + \quad H_2O(g)$
7	$Cu(\ \ ) + \quad Ag^+(aq) \rightarrow \quad Cu^{2+}(aq) + \quad Ag(\ \ )$
8	$Cu(\ \ ) + \quad O_2(\ \ ) \rightarrow \quad CuO(\ \ )$
9	$KBrO_3(s) \rightarrow \quad KBr(s) + \quad O_2(\ \ )$
10	$Ca(\ \ ) + \quad N_2(\ \ ) \rightarrow \quad Ca_3N_2(\ \ )$
11	$H^+(aq) + \quad Zn(OH)_2(s) \rightarrow \quad Zn^{2+}(aq) + \quad H_2O(\ \ )$
12	$C_2H_4(g) + \quad O_2(\ \ ) \rightarrow \quad H_2O(g) + \quad CO(\ \ ) + \quad CO_2(\ \ )$
13	$CO_2(\ \ ) + \quad Ba^{2+}(aq) + \quad OH^-(aq) \rightarrow \quad BaCO_3(s) + \quad H_2O(\ \ )$
14	$Rb(s) + \quad H_2O(l) \rightarrow \quad Rb^+(aq) + \quad OH^-(aq) + \quad H_2(\ \ )$
15	$C_7H_{14}(g) + \quad O_2(\ \ ) \rightarrow \quad CO_2(\ \ ) + \quad H_2O(l)$

Balancing chemical equations

For each of the following described reactions, write a balanced equation.

No.	
16	In an aluminium–air battery that is ideally suited to powering electrical vehicles, aluminium reacts with oxygen and water to form solid aluminium hydroxide.
17	Sulfur and steam are formed when a mixture of the gases hydrogen sulfide and sulfur dioxide are heated.
18	The first stage in the manufacturing of nitric acid involves reacting ammonia with oxygen in the presence of a catalyst. The products formed are nitrogen monoxide and steam.
19	The early Apollo moon landings used methylhydrazine (NH_2NHCH_3) as the fuel in the landing rockets, with dinitrogen tetroxide as the oxidising agent. The production of the gases, nitrogen, steam and carbon dioxide, propelled the rockets forward.
20	Iodine occurs as iodate ions, IO_3^- in nitrate deposits in Chile. To extract the iodine, the iodate ions are reacted with sulfur dioxide and water. Sulfate ions, hydrogen ions and iodide ions are formed in this reaction.

Worksheet 9.2 Precipitation reactions	

NAME: **CLASS:**

INTRODUCTION

In a precipitation reaction, an insoluble solid is formed when two solutions are mixed. An ionic equation is usually written to best represent what happen in a precipitation reaction.

An ionic equation shows only reacting species. ‘Spectator’ (non-reactive) ions are omitted. For example, when a dilute solution of barium nitrate is added to a dilute solution of sodium sulfate, a white precipitate forms. This reaction is represented by the following ‘molecular’ equation:

$$Ba(NO_3)_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaNO_3(aq)$$

Separating the ions in the soluble salts and listing all species gives:

$$Ba^{2+}(aq) + NO_3^{-}(aq) + Na^{+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s) + Na^{+}(aq) + NO_3^{-}(aq)$$

Cancelling what is common to both sides will yield the ionic equation for this reaction:

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$$

Thus, Na^+ and NO_3^- are spectator ions.

No.	Question	Answer
1	The following pairs of solutions are mixed in a test tube. Predict whether a precipitate will form and describe what you would expect to see happening in the test tube. **a** sodium carbonate + magnesium chloride **b** copper nitrate + potassium phosphate **c** zinc bromide + calcium nitrate	
2	A potassium sulfide solution is added to a iron(III) nitrate solution and a precipitate forms. **a** What is the name of this precipitate? **b** Write an ionic equation for the formation of this precipitate.	

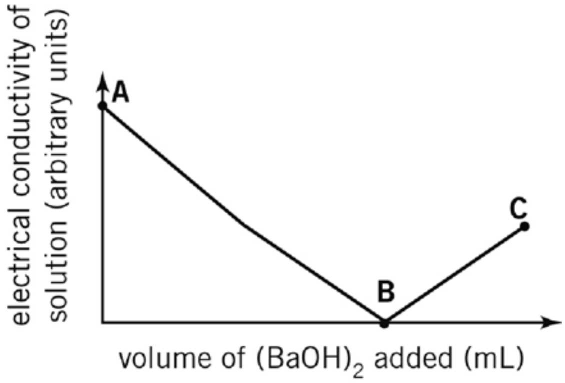
3	Write balanced ionic equations for the following reactions and identify the spectator ions in each case. **a** Barium nitrate solution is added to sodium sulfate solution **b** Sodium iodide solution is added to lead(II) nitrate solution **c** Cobalt nitrate solution is added to potassium hydroxide solution.	
4	When a sodium sulfate solution is added to a magnesium nitrate solution, no change in the two colourless solutions is observed. Explain this observation.	
5	**a** Name *two* different combinations of solutions that would produce a precipitate of calcium carbonate. **b** Write an ionic equation for the precipitation of calcium carbonate.	
6	**a** When a solution of potassium sulfate is added to some river water, a white precipitate forms. Name two positive ions that may have been in the river water? **b** Describe a further test that could be carried out on the river water to determine which of the two ions you have chosen is actually present in the water.	
7	Name two solutions you could mix together to form a green precipitate.	

Worksheet 9.3 Ions in solution	

NAME: **CLASS:**

INTRODUCTION

An experiment was conducted to investigate the reaction between solutions of sulfuric acid (H_2SO_4) and barium hydroxide ($Ba(OH)_2$). The 0.20 mol L^{-1} $Ba(OH)_2$ solution was slowly added to 20.0 mL of the 0.15 mol L^{-1} H_2SO_4 solution, and the changes in electrical conductivity of the acid solution were recorded. The results are shown in the graph below.



No.	Question	Answer
1	Write a balanced ionic equation for the reaction between H_2SO_4(aq) and $Ba(OH)_2$(aq).	
2	Name any spectator ions in the reaction between H_2SO_4(aq) and $Ba(OH)_2$(aq).	
3	What must a solution contain in order for it to conduct electricity?	
4	At which point (A, B or C) shown on the graph is there the: **a** greatest number of ions present in the solution? **b** least number of ions present in the solution?	

5	Explain why the conductivity of the solution decreases between points A and B on the graph.	
6	Explain why the conductivity is at its minimum value at point B.	
7	Explain why the conductivity of the solution increases between points B and C on the graph.	
8	What volume of $Ba(OH)_2$ solution had been added at point B on the graph? Include any working used to obtain your answer.	
9	Would you expect the same type of results graph to be obtained if the experiment was repeated using solutions of 0.15 mol L^{-1} H_2SO_4 and 0.20 mol L^{-1} NaOH? Explain your answer.	

Worksheet 9.4 Introductory stoichiometry	

NAME: **CLASS:**

INTRODUCTION

This worksheet covers introductory stoichiometric problems that involve the determination of masses and volumes of reactants and products. Several relationships that are useful for these calculations are: $\boldsymbol{n = \frac{m}{M}}$ where m is the mass in g of the substance and M is its molar mass

$\boldsymbol{c = \frac{n}{V}}$ where c is the concentration, in mol L^{-1}, of a solution and V the volume of the solution in L

$\boldsymbol{n = \frac{V}{22.41}}$ where V is the volume of a gas at STP.

No.	Question	Answer
1	Methane burns readily in oxygen to produce carbon dioxide and water according to the equation: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ What amount in mol of H_2O will be produced by the complete oxidation of 4.50 mol of CH_4 in excess oxygen?	
2	When an electric current is passed through molten sodium chloride, sodium metal and chlorine gas are produced. The equation for the reaction is: $2NaCl(l) \rightarrow 2Na(l) + Cl_2(g)$ What mass of chlorine gas would be produced from 5.00×10^2 mol of molten sodium chloride?	
3	When added to a solution of copper nitrate, what volume of 0.100 mol L^{-1} NaOH solution would be required to form 0.0720 mol of $Cu(OH)_2$? $Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$	

4	Copper metal reacts with 7 mol L^{-1} nitric acid to produce nitrogen monoxide, among other products. The ionic equation for this reaction is: $3Cu(s) + 2NO_3^-(aq) + 8H^+(aq) \rightarrow 3Cu^{2+}(aq) + 2NO(g) + 4H_2O(l)$ Determine the amount in mol of copper required to produce 215 L, measured at STP, of nitrogen monoxide gas.	
5	If aluminium filings are placed into a solution of copper(II) sulfate, copper metal is displaced and a solution of aluminium sulfate results. The equation for this redox reaction is: $2Al(s) + 3Cu^{2+}(aq) \rightarrow 3Cu(s) + 2Al^+ (aq)$ What mass of copper metal will be displaced if 5.49 g of aluminium metal is placed into a large volume of copper(II) sulfate solution?	
6	Phosphoric acid (H_3PO_4) can be generated by the oxidation of phosphorus with nitric acid according to the equation: $P(s) + 5HNO_3(aq) \rightarrow H_3PO_4(aq) + H_2O(l) + 5NO_2(g)$ If sufficient reactants are employed to produce 1.00 kg of phosphoric acid, what volume, measured at STP, of nitrogen dioxide will also be generated in the reaction?	

7	When acetic acid (CH_3COOH) is added to solid sodium carbonate, the products formed are carbon dioxide, water and a solution of sodium acetate. **a** Write a balanced equation for this reaction. **b** Calculate the mass of sodium carbonate required to react with 120 mL of a 0.156 mol L^{-1} solution of acetic acid.	
8	Ethane gas, C_2H_6, burns to form carbon dioxide and steam. **a** When a substance burns it reacts with oxygen from the air. Use this information to write an equation for the burning of ethane. **b** What amount in mol of carbon dioxide would form from the burning of 0.200 mol of ethane? **c** What volume of oxygen would be required to react with 4.30 L of ethane, assuming that the volumes are measured at the same temperature and pressure?	

Worksheet 10.1
Thermochemical equations

NAME: **CLASS:**

INTRODUCTION

Thermochemical equations are those that include an enthalpy value, ΔH. These equations can be used to infer information about particular reactions.

No.	Question	Answer
1	For each of the following changes, determine whether the process is endothermic or exothermic: **a** $CaCl_2.6H_2O(s) \xrightarrow{H_2O} CaCl_2(aq)$ $\Delta H = +18.0$ kJ **b** $4NH_3(g) + 3O_2(g) \rightarrow 2N_2(g) + 6H_2O(g)$ $\Delta H = -1250$ kJ **c** $NaCl(s) \rightarrow NaCl(l)$ **d** $I_2(g) \rightarrow I_2(s)$ **e** $2C(s) + O_2(g) \rightarrow 2CO(g)$ **f** $6CO_2(g) + 6H_2O(l) \rightarrow C_6H_{12}O_6(aq) + 6O_2(g)$	
2	For the reactions shown in questions **1a** and **1b** would you expect the temperature of the surroundings to increase or decrease during this reaction?	
3	Given the thermochemical equation: $SO_2(g) \rightarrow S(s) + O_2(g)$ $\Delta H = +297$ kJ **a** Draw an energy profile diagram for this reaction, showing the relative energies of the reactant and products. **b** Comment on the energies involved in bond-breaking and bond-formation processes for this reaction.	
4	Given the thermochemical equation $2NO_2(g) \rightarrow N_2(g) + 2O_2(g)$ $\Delta H = -67.4$ kJ determine ΔH for the following: **a** $4NO_2(g) \rightarrow 2N_2(g) + 4O_2(g)$ **b** $\frac{1}{2}N_2(g) + O_2(g) \rightarrow NO_2(g)$	

Worksheet 10.1 Thermochemical equations	

No.	Question	Answer
5	Given the equation $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ $\Delta H = -92.3$ kJ, calculate the amount of energy that would be released when 5.00 mol of hydrogen gas reacts completely with an excess of nitrogen gas.	
6	Propane burns readily in air according to the thermochemical equation $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + H_2O(l)$ $\Delta H = -2220$ kJ mol^{-1} Calculate the mass of propane required to produce 1.530 MJ of energy.	

Complete the gaps in the following sentences.

7	**a** All exothermic reactions result in a/an .... in temperature of the surroundings. **b** In an exothermic reaction, the enthalpy of the reactants is .... than the enthalpy of the products. **c** During an endothermic reaction, energy is .... from the surroundings. **d** In an .... reaction, some of the chemical potential energy stored in chemicals is converted to heat energy. **e** The breaking of a chemical bond is a process that .... energy. **f** In the formation of a chemical bond, energy is .... **g** Melting a solid to form a liquid is an .... process. **h** When steam condenses to form water, heat energy is .... to/from the surroundings. **i** The potential energy of a substance in the solid state is .... than the potential of the substance in its liquid state. **j** For an endothermic reaction, □H has a .... value. **k** The value and sign of the □H for a reaction can be calculated using the following formula: ΔH = enthalpy of products .... enthalpy of reactants.

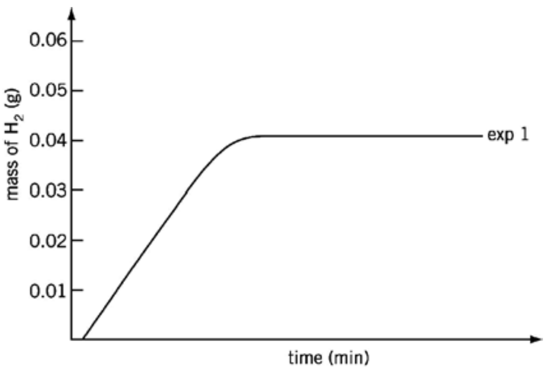
Worksheet 11.1 Rate of reaction	

Worksheet 11.1
Rate of reaction

NAME: **CLASS:**

INTRODUCTION
These questions look at the factors that affect the rate of a chemical reaction.

No.	Question	Answer
1	According to collision theory, what two criteria must be met before a molecular collision will result in a chemical reaction?	
2	When a piece of sodium is added to water, an immediate reaction occurs and sufficient heat is produced to melt the sodium and ignite the hydrogen produced in the reaction. A solution of sodium hydroxide is the other product formed in the reaction. **a** Write an equation for the reaction. **b** Is this reaction exothermic or endothermic? **c** Does the reaction have a large or small activation energy? **d** Draw an energy profile graph for the reaction, labelling the reactants, products, enthalpy change and activation energy.	
3	When a strip of magnesium is placed into a solution of 0.1 mol L^{-1} hydrochloric acid, hydrogen gas only slowly evolves. When the same experiment is performed with 2 mol L^{-1} hydrochloric acid, the evolution of gas is substantial and dramatic. Explain this observation in terms of the collision theory	

No.	Question	Answer
4	Explain each of the following observations in terms of the factors that affect rate of reaction and the collision theory. **a** Sulfuric acid is very reactive when it is hot and concentrated. **b** Petrol in its vapour form is far more flammable than in its liquid form. **c** Food takes longer to decay when it is placed in the refrigerator.	
5	Hydrogen may be prepared by reacting magnesium and hydrochloric acid. Three experiments were conducted to investigate this reaction. In each experiment 50.0 mL of 1.0 mol L^{-1} HCl was used. The mass of hydrogen evolved was measured at 5 minute intervals. • Experiment 1 was performed at a temperature of 25°C, using 0.5 g of magnesium ribbon. • Experiment 2 was performed at a temperature of 40°C, using 0.5 g of magnesium ribbon. • Experiment 3 was performed at a temperature of 25°C, using 0.3 g of magnesium powder. Results for experiment 1 are shown in the graph below. On the same axes sketch the expected results for experiment 2 and experiment 3. 	

Worksheet 11.2 The role of catalysts	

NAME: **CLASS:**

INTRODUCTION

Catalysts perform an important role in increasing the rate of chemical reactions. These questions look at classes of catalysts and how they perform their function.

No.	Question	Answer
1	**a** Draw an energy profile diagram for the reaction $A \rightarrow B + C$ for which $\Delta H = -150$ kJ. The enthalpy of $(B + C)$ can be taken as zero, and the activation energy for the forward reaction is 100 kJ. **b** Determine the activation energy and enthalpy change of the reverse reaction $B + C \rightarrow A$. **c** On the same diagram, use a dotted line to represent the activation energy of the reaction if a catalyst were to be used.	
2	Many solid catalysts are used in pellet form or as wire gauze. What is the advantage of producing catalysts in these forms rather than as a large lump?	
3	Hundreds of thousands of chemical reactions occur within the body's cells every second. What is the name given to the biological catalysts that ensure these reactions occur very quickly?	

No.	Question	Answer
4	Biological catalysts can be many millions of times more efficient that industrial catalysts and yet they are rarely used in industrial manufacturing processes. Why not?	
5	Catalysts ‘lower the activation energy without actually being consumed in the chemical reaction’. Explain how catalysts achieve these two seemingly contradictory roles.	

Worksheet 12.1 Stoichiometry	

NAME: **CLASS:**

INTRODUCTION

Stoichiometric problems can be tackled by remembering five basic steps.

Step 1: Write a balanced chemical equation for the reaction.

Step 2: List all data given, including relevant units. Remember to also write down the symbol of the unknown quantity.

Step 3: Convert the data given to amount in mole, using the relevant formulas:

$$n = \frac{m}{M} \qquad n = c \times V \qquad n = \frac{N}{N_A} \qquad n = \frac{V}{V_M}$$

Step 4: Use the chemical equation to determine the mole ratio of the unknown quantity to the known quantity. This ratio enables calculation of the amount in mole of the unknown quantity.

Step 5: Finally, convert this amount in mole back into the relevant units of the unknown.

In all calculation questions, take care to show all steps in your working, including reacting ratios where relevant. Also, take care to add the correct unit to your answer (e.g. mol, L) and give your answer to the correct number of significant figures.

No.	Question	Answer
1	Sulfur dioxide in the atmosphere contributes to acid rain. The equation for formation of the acid is represented by the equation: $2SO_2(g) + O_2(g) + 2H_2O(l) \rightarrow 2H_2SO_4(aq)$ What mass of sulfuric acid will form from 50.0 L (measured at STP) of sulfur dioxide?	

No.	Question	Answer
2	23.8 mL of hydrochloric acid is just neutralised by 29.9 mL of 2.86 mol L^{-1} sodium hydrogencarbonate solution, according to the equation: $HCO_3^-(aq) + H^+(aq) \rightarrow CO_2(g) + H_2O(l)$ Determine the concentration, in mol L^{-1}, of the hydrochloric acid solution used.	
3	In a car accident, the impact triggers ignition of a detonator cap in the air bag, which causes sodium azide (NaN_3) to decompose explosively. **a** Write the equation to show the decomposition of sodium azide into solid sodium and nitrogen gas. **b** If the bag contained 75 g of sodium azide, what volume of gas, measured at 0°C and 101.3 kPa, would form?	
4	Sodium thiosulfate reacts with bromine in alkaline solution according to the equation: $Na_2S_2O_3(aq) + 4Br_2(l) + 10NaOH(aq) \rightarrow 2Na_2SO_4(aq) + 8NaBr(aq) + 5H_2O(l)$ In order to completely react with 10.0 mL of liquid bromine of density 3.12 g mL^{-1}, 239 mL of NaOH solution was added along with excess $Na_2S_2O_3$. Determine the concentration, in mol L^{-1}, of the sodium hydroxide solution required.	

No.	Question	Answer
5	Arsenic undergoes oxidation by a hot, concentrated solution of sodium hydroxide to produce sodium arsenate and hydrogen gas according to the equation: $2As(s) + 6OH^{-}(aq) \rightarrow 2AsO_3^{3-}(s) + 3H_2(g)$ 6.57 g of arsenic is reacted with 250 mL of 0.779 mol l^{-1} sodium hydroxide solution. Calculate **a** the mass of hydrogen gas evolved in the process **b** the concentration, in mol L^{-1} of the sodium arsenate solution that remains after the reaction.	

Worksheet 12.2 Solution stoichiometry	

NAME: **CLASS:**

INTRODUCTION

These problems involve stoichiometric calculations where one or more of the reactants or products are in solution.

No.	Question	Answer
1	Determine the amount in mol of barium sulfate that will be precipitated when 200.0 mL of 0.450 mol L^{-1} barium nitrate solution is added to an excess of sodium sulfate solution, given that the equation for the reaction is: $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$	
2	Liquid hydrazine reacts explosively with hydrogen peroxide according to the equation: $N_2H_4(l) + 2H_2O_2(aq) \rightarrow N_2(g) + 4H_2O(g)$ Determine the volume, at STP, of gaseous products that would be formed by the reaction of 3.50 L of 4.30 mol L^{-1} hydrogen peroxide with excess hydrazine.	

3	When sulfur dioxide gas is bubbled through a solution of sodium hydroxide, a solution of sodium sulfite is produced according to the equation: $SO_2(g) + 2OH^-(aq) \rightarrow SO_3^{2-}(aq) + H_2O(l)$ What volume of 0.638 mol L^{-1} NaOH solution will be required to absorb exactly 70.9 g of sulfur dioxide gas?	
4	Mercury reacts with concentrated nitric acid to produce nitrogen monoxide according to the equation: $3Hg(l) + 8HNO_3(aq) \rightarrow Hg(NO_3)_2(aq) + 2NO(g) + 4H_2O(l)$ What mass of mercury will react exactly with 125 mL of 14.0 mol L^{-1} nitric acid?	
5	What volume of 0.0712 mol L^{-1} ammonium bromide solution will be required to precipitate all of the mercury(II) ions from 33.6 mL of 0.144 mol L^{-1} mercury(II) nitrate solution? The ionic equation for the reaction is: $Hg^{2+}(aq) + 2Br^-(aq) \rightarrow HgBr_2(s)$	
6	A 0.600 g impure sample of sodium carbonate required 18.85 mL of 0.500 mol L^{-1} hydrochloric acid for complete reaction according to the equation: $2H^+(aq) + Na_2CO_3(s) \rightarrow CO_2(g) + H_2O(l) + 2Na^+(aq)$ Calculate the percentage purity, by mass, of the sodium carbonate, assuming that the impurity did not react with acid.	

7	A sample of commercial vinegar contains 4.58% acetic acid by mass. A 14.72 g sample of vinegar was diluted to 100.0 mL with water, and 25.00 mL of 0.307 mol L^{-1} sodium hydroxide solution was then added. Calculate the concentration, in mol L^{-1}, of acetate ion in the final solution. $CH_3COOH(aq) + OH^-(aq) \rightarrow CH_3COO^-(aq) + H_2O(l)$	

Worksheet 12.3	
Excess reagent stoichiometry	

NAME: **CLASS:**

INTRODUCTION

This set of stoichiometry questions includes problems that involve excess and limiting reactants. The calculation of the limiting reagent must be shown in the answers to these questions.

No.	Question	Answer
1	Ethene (C_2H_4) burns readily in oxygen to produce carbon dioxide and water. A sealed container is filled with 2.0 mol of ethene and 9.0 mol of oxygen gas and the gaseous mixture is ignited. **a** Write an equation, including states, for this combustion reaction. **b** Which gas is the limiting reagent? **c** What amount in mol of carbon dioxide will be formed in the reaction?	
2	Sodium metal reacts vigorously with water to produce a solution of sodium hydroxide according to the equation: $2Na(s) + 2H_2O(l) \rightarrow 2Na^+(aq) + 2OH^-(aq) + H_2(g)$ Determine the concentration of sodium hydroxide solution that would result from the complete reaction of 0.835 g of sodium in 180 mL of water. (Assume the density of water at room temperature is 1 g mL^{-1})	
3	10.0 g of chromium(III) oxide is dissolved in 120 mL of 0.774 mol L^{-1} potassium bromate ($KBrO_3$) solution according to the equation: $5Cr_2O_3(s) + 6BrO_3^-(aq) + 4OH^-(aq) \rightarrow 5Cr_2O_7^{2-}(aq) + 2H_2O(l) + 3Br_2(aq)$ Determine which reagent is in excess and thus calculate the mass of bromine formed in the reaction.	

4	45.5 mL of 0.968 mol L^{-1} tin(II) chloride solution is added to 58.2 mL of 1.223 mol L^{-1} ammonium hydroxide solution to generate a precipitate of tin(II) hydroxide. **a** Write an equation, including states, for this precipitation reaction. **b** What mass of precipitate is produced?	

Worksheet 12.4 Empirical and molecular formula calculations	

NAME: **CLASS:**

INTRODUCTION

The **empirical formula** of a compound is defined as the simplest whole-number ratio of atoms of different elements in the compound.
The **molecular formula** of a compound is defined as the actual number of atoms of different elements covalently bonded in a molecule, and is a whole-number multiple of the empirical formula.
The **percentage composition** of a compound can be determined from its formula or from experimental mass proportion data.

No.	Question	Answer
1	Titanium oxide, TiO_2, is widely used as a pigment in high-quality artists' paints due to its brilliant white colour. Determine the percentage composition of TiO_2.	
2	An oxide of sulfur contains 60% oxygen. Determine the empirical formula of the oxide.	
3	A group 1 metal chloride contains 47.6% chlorine. What is the formula of this ionic compound?	
4	Write empirical formulas for each of the following compounds: **a** CH_4 **b** N_2H_4 **c** $C_2H_4O_2$	

5	A molecule has an empirical formula C_2H_4O and a relative molecular mass of 88. What is its molecular formula?	
6	0.300 mole of a sample of a hydrocarbon is found to have a mass of 24.6 g. If the empirical formula of the compound is C_3H_5, determine its molecular formula.	
7	A student is given a 2.486 g sample of a purple crystalline solid. Upon analysis it is found to consist of potassium (0.614 g), manganese (0.863 g) and oxygen (1.006 g). Determine the empirical formula of the compound.	
8	A compound undergoes analysis to establish the following elemental composition: carbon: 40.0% hydrogen: 6.67% oxygen: 53.3% If the compound, known as a carbohydrate, has a molecular mass of approximately 60, determine both its empirical and molecular formulas.	

Worksheet 12.5 Determining the molecular formula of a gas	

NAME: **CLASS:**

INTRODUCTION

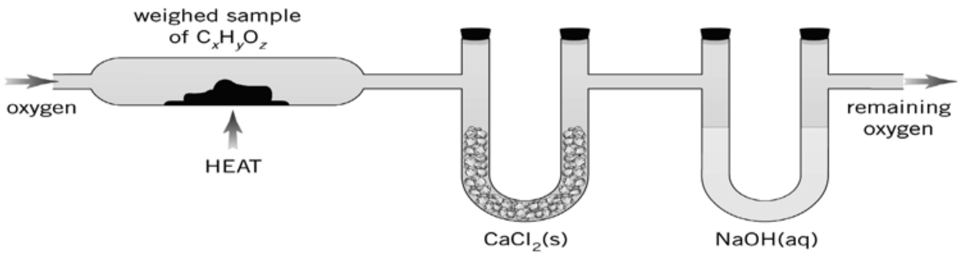
Empirical formula: The formula showing the smallest whole-number ratio of atoms in a substance.
Molecular formula: The formula showing the actual number of atoms of each element present in a molecule of a compound. For example, the molecular formula of glucose is $C_6H_{12}O_6$ and its empirical formula is CH_2O.

No.	Question	Answer
1	A compound used in ceramics contains, by mass, 22.8% Na, 21.5% B and 55.7% O. What is its empirical formula?	
2	A sweet-smelling organic compound contains 0.0556 mol of carbon, 0.112 g of hydrogen atoms and 9.57×10^{21} oxygen atoms. Calculate its empirical formula.	

Worksheet 12.5 Determining the molecular formula of a gas	

No.	Question	Answer
3	If a compound contains 75.7% arsenic and 24.3% oxygen by mass, and has a molar mass of 395.6 g mol^{-1}, what is its molecular formula?	

The apparatus shown below may be used to determine the empirical and molecular formulas of an unknown compound of general formula $C_xH_yO_z$.



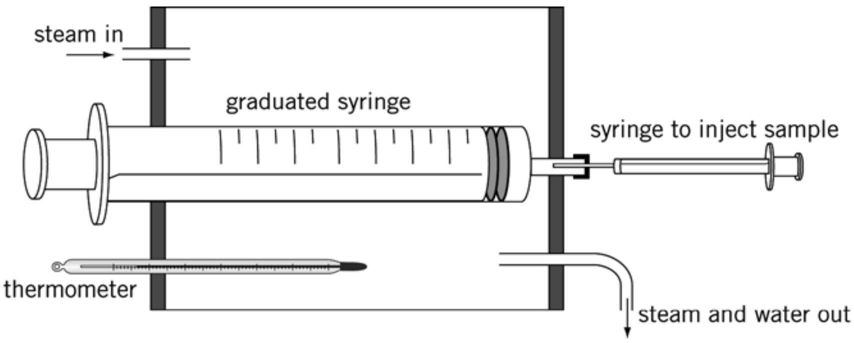
The questions that follow refer to this apparatus.

No.	Question	Answer
4	What is the purpose of the $CaCl_2(s)$?	
5	What is the purpose of the NaOH(aq)?	

Worksheet 12.5	
Determining the molecular formula of a gas	

No.	Question	Answer
6	14.34 g of a hydrocarbon was burned in excess oxygen. The mass of the NaOH(aq) increased by 43.54 g, while the mass of $CaCl_2$(s) increased by 22.26 g. What is the empirical formula of the hydrocarbon?	

A method used to determine the relative molecular mass of a volatile liquid is shown below.



Steam is passed through the outer jacket surrounding a graduated syringe that contains a small, measured volume of air. When the temperature has stabilised, a weighed sample of a few mL of liquid is injected into the graduated syringe, using a small hypodermic syringe. The liquid vaporises and the final volume of air plus sample is recorded. The formula PV = 8.315 nT, where the P is in kPa, V in L and T in K, can then be used to determine the amount in mol of gas in the syringe.

The questions which follow refer to this apparatus.

No.	Question	Answer
7	Why would this method be unsuitable for a liquid with a boiling point above about 90°C?	

Worksheet 12.5

Determining the molecular formula of a gas

No.	Question	Answer
8	Using the apparatus shown, 0.16 g of liquid was vaporised to produce a volume of 46 mL, at a temperature of 100°C and a pressure of 101.3 kPa. Determine the molar mass of the liquid.	

Worksheet 12.5 Determining the molecular formula of a gas	

No.	Question	Answer
9	In one experiment an 18.99 g sample of the compound was burnt in excess oxygen. When the gases evolved were passed through anhydrous $CaCl_2$, its mass increased by 11.38 g. The remaining gases, when bubbled through a NaOH solution, increased its mass by 27.83 g. In a separate experiment, a 6.21 g sample of the compound was vaporised. The vapour occupied 2.17 L at 200°C and 125 kPa. Calculate the molecular formula of the compound.	

Worksheet 12.6 Gravimetric analysis of a fertiliser	

NAME: **CLASS:**

INTRODUCTION

The purity of a sample of ammonium sulfate fertiliser was determined by gravimetric analysis in which the sulfate was precipitated as barium sulfate. The following steps were used in this procedure:

1 A sample of about 5 g of the fertiliser was dried in an oven for an hour at 110ºC.
2 The dried fertiliser was ground into fine powder using a mortar and pestle.
3 1.00 g of the dried, ground fertiliser was accurately weighed out into a 250 mL beaker.
4 150 mL of distilled water and 2 mL of 10 mol L^{-1} HCl were added to the fertiliser with stirring.
5 The solution was heated to nearly boiling.
6 The fertiliser solution was maintained at that temperature while the precipitating agent was prepared. If the fertiliser was pure, it was calculated that 37.8 mL of the 0.200 mol L^{-1} $BaCl_2$ stock solution would be required. 45 mL of the stock solution was placed in a 100 mL beaker and heated to nearly boiling.
7 The fertiliser solution was stirred vigorously as the hot barium chloride solution was added slowly.
8 Heating of the beaker continued while the precipitate was allowed to settle. A few drops of the supernatant liquid were tested for complete precipitation by adding them to a few drops of barium chloride solution on a watchglass. No cloudiness was observed.
9 The fertiliser–barium chloride mixture was covered and left to ‘digest’ just below boiling point for one hour. (*This digestion process is necessary for experiments involving barium sulfate because the crystals formed are initially too small to filter. Digestion allows larger crystals to form*.)
10 The hot solution was filtered under vacuum filtration into a pre-weighed sintered glass crucible.
11 The precipitate was washed several times with hot water.
12 The crucible and precipitate were dried to constant mass in an oven at 60ºC over two successive days; 1.63 g of precipitate was obtained.

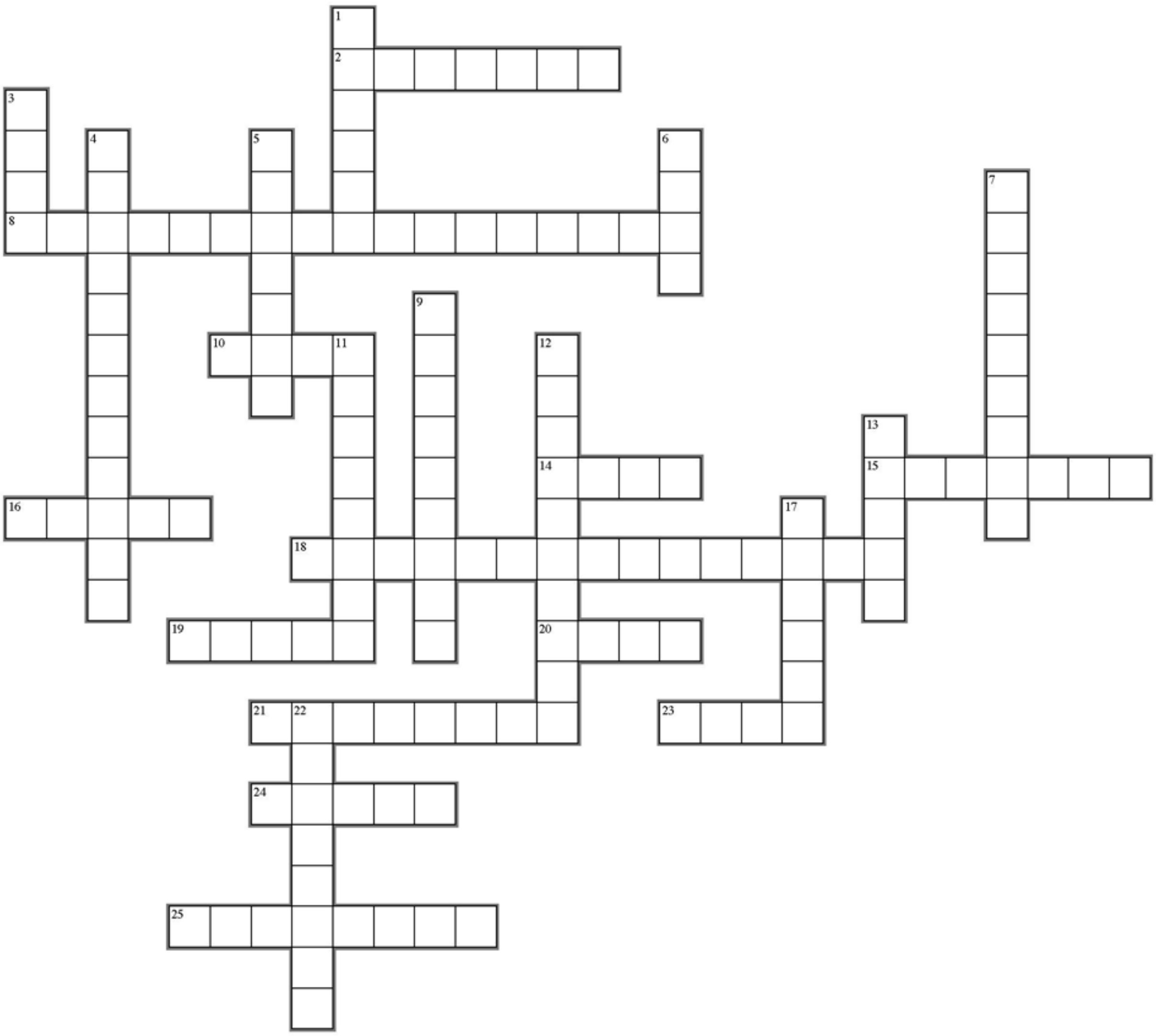
No.	Question	Answer
1	Why was it necessary to dry and grind the fertiliser before weighing?	
2	If the fertiliser was 100% ammonium sulfate, what amount in mol of ammonium sulfate would be present in 1.00 g?	

Worksheet 12.6 Gravimetric analysis of a fertiliser	

No.	Question	Answer
3	This procedure assumes that there were no insoluble components in the fertiliser. If there had been, what extra step would have been necessary?	
4	Step 6 states that 37.8 mL of barium chloride was needed. Show how this volume was calculated.	
5	If 37.8 mL was the maximum volume needed, why was 45 mL used?	
6	How was it shown that complete precipitation had taken place?	
7	Why was the precipitate washed in hot water?	
8	Define the term ‘weighed to constant mass’.	
9	Use the final mass of precipitate to calculate the % m/m of ammonium sulfate in the fertiliser.	
10	Where could errors have occurred in this experiment?	

Worksheet 13.1
Common acids and bases

NAME: **CLASS:**



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Common acids and bases

ACROSS

2. Many window cleaners contain this basic substance.
8. The sodium salt of this ion is baking soda.
10. A common name for a hydroxide used to neutralise soil acidity.
14. The taste of an acid.
15. The analgesic chemically known as acetyl salicylic.
16. ______ acid is a mild antiseptic used in ointments for dressing wounds.
18. A caustic chemical used in drain cleaners. (6, 9)
19. ______ acid is found in grapes and apples, particularly unripe ones.
20. Carbonic acid is found naturally in this watery substance.
21. _______ acid is Vitamin C.
23. Many watch batteries have this hydroxide as a reactant.
24. Hydrofluoric acid is used to etch this substance.
25. Produced when hydrochloric acid reacts with iron.

DOWN

1. ______ acid is found in wine and tea.
3. The chemical formula for the carboxylic acid functional group.
4. ______ acid is the main component of concrete cleaner.
5. A fatty acid with molecular formula of $C_{18}H_{36}O_2$, found widely in solid fats.
6. Human urine contains this basic substance.
7. _______ acid is the systematic name for the acid present in bee stings.
9. One of the hydroxides commonly used in antacid preparations.
11. ______ acid is an alternative name for the acid in vinegar.
12. ______ acid is an active ingredient in Coca-Cola.
13. One of the products of the reaction between an acid and a metal hydroxide.
17. Lemons and oranges contain ______ acid.
22. ______ acid is found in most car batteries.

Worksheet 13.2 Revision of acids and bases	

NAME: **CLASS:**

INTRODUCTION

These reactions are typical of acids:

acid + metal → salt + hydrogen gas (a redox reaction)
acid + metal hydroxide → salt + water
acid + metal oxide → salt + water
acid + metal carbonate → salt + water + carbon dioxide
acid + metal hydrogencarbonate → salt + water + carbon dioxide

According to the Brønsted-Lowry theory:
Acids are proton (H^+) donors; bases are proton acceptors. Acid–base reactions involve the transfer of protons (H^+ ions). Acidic species may be described as monoprotic (donate one proton), diprotic (donate two protons), triprotic (donate three protons) or amphiprotic (amphoteric) (both donate and accept protons).

Acid–base strength is a measure of how readily protons are donated or accepted.

The pH scale is used as a measure of the acidity or basicity of a solution. The scale is usually applied over the range 0 to 14 (but does extend beyond these values). At 25°C, acidic solutions have a pH less than 7, basic solutions have a pH greater than 7 and neutral solutions have a pH equal to 7.

No.	Question	Answer
1	Write balanced ionic equations for the reactions that occur when the following chemicals are mixed: **a** Solid potassium hydrogencarbonate and sulfuric acid **b** Iron(III) oxide and nitric acid **c** Calcium hydroxide solution and hydrochloric acid	
2	Write balanced equations to illustrate the following reactions: **a** Dissociation of $Ca(OH)_2$ in aqueous solution **b** Successive ionisations of H_2SO_4 **c** Neutralisation of $Ba(OH)_2$ solution with HCl solution	

Worksheet 13.2

Revision of acids and bases

No.	Question	Answer
3	Write the formulas of: **a** the conjugate bases of HS^- and OH^- **b** the conjugate acids of HSO_4^- and $H_2PO_4^-$.	
4	For each of the following, identify the Brønsted-Lowry conjugate acid–base pairs involved in the reaction: **a** $CO_3^{2-}(aq) + H_3O^+(aq) \rightarrow H_2O(l) + HCO_3^-(aq)$ **b** $HS^-(aq) + OH^-(aq) \rightarrow S^{2-}(aq) + H_2O(l)$ **c** $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$	
5	HSO_4^- is an amphiprotic (amphoteric) ion. Write chemical equations to show this ion acting as: **a** an acid **b** a base.	
6	Give two reasons why two acid solutions of equal concentration could have different pH values.	
7	Give an explanation for each of the following observations: **a** The electrical conductivity of a 1.0 mol L^{-1} solution of methanoic acid (HCOOH) is less than that of a 1.0 mol L^{-1} solution of hydrochloric acid (HCl). **b** Acetic acid (CH_3COOH) is monoprotic, even though it contains four hydrogen atoms.	

Revision of acids and bases

No.	Question	Answer
8	List the following 1.0 mol L^{-1} solutions in order of decreasing pH. Give reasons for your order. $NaOH$, H_2O, NH_3, CH_3COOH, H_2SO_4, HNO_3	
9	20.0 mL of a solution of pH 3.0 is diluted to produce a total volume of 200.0 mL. What is the pH of the resulting solution?	
10	Write equations for the following: (if no reaction occurs write ‘no reaction’) **a** carbon dioxide is dissolved in water **b** sulfur dioxide is bubbled through a solution of sodium hydroxide **c** sodium oxide is added to a solution of potassium hydroxide **d** aluminium oxide is added to a solution of sodium hydroxide	

Worksheet 14.1
Oxidation numbers and redox equations

NAME: **CLASS:**

INTRODUCTION

The concept of oxidation numbers (ON) or oxidation states was applied to determine whether or not electrons had moved from one species to another in a chemical reaction. An oxidation–reduction (redox) reaction is one in which one or more atoms change oxidation numbers. Oxidation occurs when an atom's oxidation state becomes more positive, indicating that electrons have been lost. Reduction occurs when an atom's oxidation state becomes less positive, indicating that electrons have been gained.

OXIDATION →

… –5 –4 –3 –2 –1 0 +1 +2 +3 + 4 +5 …

← REDUCTION

The oxidation numbers assigned to atoms in covalent compounds are hypothetical charges, and the atoms do not really have these charges within the compound, since they are only sharing electrons. For example, in the compound CO_2 we say that carbon has a +4 ON and each oxygen has a –2 ON, but the atoms do not really have these charges.

Recall that the **oxidant** is the reactant being reduced, so its ON will decrease. The **reductant** is the reactant being oxidised, so its ON will increase.

Rules for assigning oxidation numbers

1 The oxidation number of an element is zero.
2 For a monatomic ion, the oxidation number is the charge on the ion.
3 The oxidation number of combined hydrogen is usually +1.
4 The oxidation number of combined oxygen is usually –2.
5 The sum of all oxidation numbers of atoms in a compound is zero.
6 The sum of all oxidation numbers of atoms in an ion is equal to the charge on that ion.

No.	Question	Answer
1	Write the formula of a substance in which nitrogen has the following oxidation numbers: **a** +3 **b** +5 **c** 0 **d** –3	

Worksheet 14.1 Oxidation numbers and redox equations	

No.	Question	Answer
2	What is the oxidation number of the atom in bold type in each of the following? **a** $H_2\mathbf{S}O_4$ **b** $K_2\mathbf{Cr}_2O_7$ **c** $\mathbf{C}O_2$ **d** $H_2\mathbf{O}_2$	
3	For the following equations, determine which are redox processes. For those that are redox reactions, identify the oxidant and the reductant. **a** $2OH^-(aq) + Cr_2O_7^{2-}(aq) \rightarrow 2CrO_4^{2-}(aq) + H_2O(l)$ **b** $I_2O_5(s) + 5CO(g) \rightarrow I_2(s) + 5CO_2(g)$ **c** $PBr_3(l) + 3H_2O(l) \rightarrow H_3PO_3(aq) + 3HBr(aq)$ **d** $2Hg^{2+}(aq) + N_2H_4(aq) \rightarrow 2Hg(l) + N_2(g) + 4H^+(aq)$ **e** $3H_2S(g) + 2H^+(aq) + 2NO_3^-(aq) \rightarrow 3S(s) + 2NO(g) + 4H_2O(l)$ **f** $3NO_2(g) + H_2O(l) \rightarrow 2HNO_3(aq) + NO(g)$	
4	The highest positive oxidation number possible for any atom is equal to the number of electrons in its valence shell. For example, nitrogen has a maximum oxidation number of +5. What is the maximum oxidation number of: **a** phosphorus? **b** magnesium? **c** chlorine?	
5	Occasionally you will find that an atom in a compound has an oxidation number of zero. What is the oxidation number of each atom in glucose, $C_6H_{12}O_6$?	

Worksheet 14.1 Oxidation numbers and redox equations	

Rules for balancing redox half-equations in acidic media

1 Write the species undergoing oxidation or reduction on the left and its conjugate on the right, leaving out any spectator ions.
2 Balance for atoms other than hydrogen and oxygen.
3 Balance for oxygen by adding water, H_2O, to the appropriate side of the equation.
4 Balance for hydrogen ions, H^+, to the appropriate side of the equation.
5 Balance for charge by adding electrons to the side with the greater positive charge.

No.	Question	Answer
6	For each of the following, write the two half-equations, then add them together to get the overall redox equation. Make sure you leave out any spectator ions. **a** Acidified potassium dichromate solution ($K_2Cr_2O_7$) is used to oxidise ethanol (C_2H_5OH) to acetic acid. The dichromate ion is converted to chromium(III) ions. **b** The bromate ion, BrO_3^- may be used in acidic solution to oxidise iodide ions to iodine. The reduction product is the bromide ion. **c** Purple acidified potassium permanganate solution ($KMnO_4$) is decolourised by acidified iron(II) sulfate solution, producing Mn^{2+} and Fe^{3+}. **d** Hydrogen peroxide (H_2O_2) is added to acidified potassium permanganate, and oxygen is evolved (Mn^{2+} is also formed). **e** Hydrogen sulfide (H_2S) gas is bubbled into a solution of acidified potassium dichromate, producing a deposit of sulfur and Cr^{3+}.	

Worksheet 14.2

Metal reactivity and redox equations

NAME: **CLASS:**

INTRODUCTION

Metals vary in their reactivity. More reactive metals (stronger reductants) lose their valence electrons more easily. An activity series of metals can be described, where metals further up the list are more reactive than those further down the list. Part of such an activity series is shown below.

Increasing reactivity

Lithium (Li)
Potassium (K)
Barium (Ba)
Sodium (Na)
Calcium (Ca)
Magnesium (Mg)
Aluminium (Al)
Manganese (Mn)
Zinc (Zn)
Iron (Fe)
Nickel (Ni)
Lead (Pb)
Copper (Cu)
Silver (Ag)

Metals further up the list (stronger reductants) will displace a metal further down the list from a solution of its ions. For example, when manganese is added to a solution of copper(II) sulfate, the following reaction occurs:

$$Cu^{2+}(aq) + Mn(s) \rightarrow Cu(s) + Mn^{2+}(aq)$$

In this reaction, electrons are passed from the manganese atoms to the copper(II) ions.

No.	Question	Answer
1	Name two metals that are: **a** less reactive than aluminium **b** more reactive than calcium.	
2	Metal X is added to a solution of nickel nitrate and a coating of nickel metal appears on metal X. Name two metals that are less reactive than X.	
3	What would you expect to observe if a piece of manganese were to be placed in a solution of lead(II) nitrate?	

4	What would you expect to observe if a piece of nickel were to be placed in a solution of zinc nitrate?	
5	**a** Write the half equations for the reaction of solid magnesium with a solution of iron(II) sulfate. **b** Write the overall ionic equation for this reaction.	
6	Explain, with the aid of an equation, how solid lead could be produced from a solution of lead nitrate.	
7	Explain, using an appropriate equation, why a nickel spatula should not be used to stir solutions of silver nitrate.	
8	A solution of zinc nitrate is to be stored, but only four different metal containers are available. They are made of copper, nickel, aluminium and silver. Which would be the best choice?	
9	Two colourless solutions are placed side by side. One is magnesium nitrate and the other is zinc nitrate. Describe a simple test you could conduct to determine which was which.	

A student conducted an experiment to determine the activity series for metals A, B, C and D. Solid samples of each metal, as well as 0.1 mol L^{-1} solutions of their nitrate salts were used. The different metals were placed in the different solutions, and the results shown below were obtained.

	Metal A	Metal B	Metal C	Metal D
0.1 mol L^{-1} ANO_3	No reaction	Reaction	Reaction	Reaction
0.1 mol L^{-1} $B(NO_3)_2$	No reaction	No reaction	Reaction	No reaction
0.1 mol L^{-1} CNO_3	No reaction	No reaction	No reaction	No reaction
0.1 mol L^{-1} $D(NO_3)_3$	No reaction	Reaction	Reaction	No reaction

Worksheet 14.2 Metal reactivity and redox equations	

No.	Question	Answer
10	Four of these experiments were unnecessary. Which are they?	
11	Which of the metals forms ions with a charge of +3?	
12	Which of the metals could possibly belong to the same group in the periodic table as calcium?	
13	Which of the metals would form a compound with chlorine that has the general formula XCl?	
14	Write balanced ionic equations for the reactions, in aqueous solution, occurring between: **a** metal B and $D(NO_3)_3$ **b** metal C and $B(NO_3)_2$ **c** metal D and ANO_3.	
15	Based on the results shown, list the activity series for these four metals from most reactive to least reactive.	

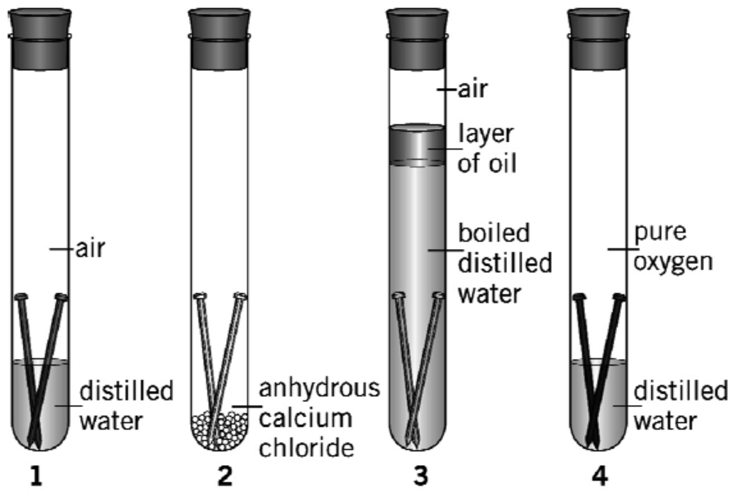
Worksheet 14.3 Corrosion of iron	

NAME: **CLASS:**

INTRODUCTION

The corrosion of iron, and subsequent rust formation, is an example of a redox reaction. This particular redox reaction is one that costs the Australian economy millions of dollars per year. In this worksheet we will consider factors that affect the rate of corrosion of iron, and some methods to prevent or slow corrosion.

The diagram below shows four tubes, each containing iron nails. Tubes were set up and left for several days. The results obtained for the experiment are given below the test tubes. Consider these results when answering the following questions.



Results: **rust on nails** **no rust** **no rust** **rust on nails**

No	Question	Answer
1	Write balanced half-equations for: **a** the oxidation of iron atoms to iron(II) ions **b** the reduction of oxygen in the presence of water to form hydroxide ions.	
2	Suggest why no corrosion has occurred in tube 2.	
3	Suggest why no corrosion has occurred in tube 3.	

4	More rust formed in tube 4 than in tube 1. Suggest a reason for this.	
5	Suggest why corrosion of an iron object is reduced (or stopped) in each of the following situations: **a** Outer space **b** At the bottom of the ocean **c** When the object is coated with oil **d** When connected with a wire to a piece of magnesium **e** When the object is connected to the negative terminal of a DC power supply (called cathodic protection).	
6	Metal rubbish bins are often coated with zinc. Metal food containers are coated with tin. **a** State two ways in which the zinc coating on rubbish bins protects the iron from corrosion. **b** State one way in which the tin coating on containers protects the iron from corroding. **c** Why is a zinc coating not suitable for use on food containers?	
7	Match each iron object to an appropriate corrosion prevention method. Gardening secateurs — Galvanising (coating with Zn) Natural gas pipeline — Cathodic protection with a DC power supply Wharf — Coating with oil Patio furniture — Attachment to sacrificial anodic magnesium blocks A shed roof — Painting	

Worksheet 15.1 Electrolysis reactions	

NAME: **CLASS:**

INTRODUCTION

Electrolysis is the process whereby electrical energy is converted into chemical potential energy. This process is used to drive non-spontaneous reactions and to manufacture a range of industrially important chemicals and metals that would otherwise be difficult to produce.

No	Question	Answer
1	The reactive metallic elements such as aluminium, sodium and potassium were not discovered until the early 19th century. What other discovery or invention was needed before these elements could be isolated from their salts?	
2	Explain why the products obtained from the electrolysis of a molten salt may be different from those of an aqueous solution of the same salt.	
3	Draw an electrolytic cell representing the electrolysis of molten lithium fluoride using inert electrodes. On or below your diagram you should clearly indicate: **a** the polarity of the electrodes **b** the anode and cathode **c** the oxidation and reduction reactions **d** the oxidant and reductant species **e** the overall equation representing the cell reaction.	
4	A molten mixture of magnesium chloride and sodium iodide is electrolysed. Determine the products that would initially be formed at the anode and the cathode. Explain your choices.	

Water can act as an oxidant or a reactant if it is present in an electrolytic cell, according to the following equations:

$2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ (acting as an oxidant)

$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$ (acting as a reductant)

Lists of oxidants and reductants, in order of decreasing strength, can be set up to include water:

Reductants	**Oxidants**
K (strongest reductant)	F_2 (strongest oxidant)
Na	Cl_2
Mg	Br_2
Zn	Ag^+
Pb	I_2
Cu	Cu^{2+}
I^-	Pb^{2+}
Ag	H_2O
H_2O	Zn^{2+}
Br^-	Mg^{2+}
Cl^-	Na^+
F^- (weakest reductant)	K^+ (weakest oxidant)

In an electrolysis cell, usually the strongest oxidant present in the cell is reduced at the cathode (negative electrode) and the strongest reductant is oxidised at the anode (positive electrode).

No	Question	Answer
5	When a 1 mol L^{-1} solution of copper bromide is electrolysed, a brown coating forms on the cathode and a colourless, odourless gas forms at the anode. **a** What substance is being formed at the cathode? **b** Write a half equation for reduction reaction that occurs at the cathode reaction. **c** What is the strongest reductant in the cell? **d** Write a half equation for the oxidation reaction that occurs at the anode. **e** What gas is being formed at the anode?	

6	**a** Determine the products formed at each inert electrode during the electrolysis of a 1 mol L^{-1} lead(II) nitrate solution (nitrate ions do not react). **b** Give half equations for the electrode reactions. Identify each as an oxidation or a reduction reaction.	
7	A student electrolyses an aqueous solution known to be either potassium chloride or zinc iodide. Bubbles of a colourless gas form at the anode of the cell during the electrolysis. **a** Which solution is being used? **b** What gas is being evolved at the anode? **c** What should be observed at the cathode of this cell? **d** Give half equations for the reactions occurring at the anode and the cathode.	
8	**a** Predict the products at the anode and cathode during the electrolysis of a solution of magnesium chloride when copper electrodes are used, giving reasons for your answers. **b** Give the half equations for the anode and cathode reactions.	

Worksheet 15.2 Redox chemistry definitions	

NAME: **CLASS:**

INTRODUCTION

Application of redox chemistry to electrolytic cells requires a clear understanding of the meaning of key terms. This worksheet is designed to revise the definition of terms associated with redox reactions and their application in electrolytic cells.

No.	Question	Answer
1	What does the acronym 'OIL RIG' represent in relation to redox chemistry?	
2	**a** Explain how oxidation numbers can be used to determine whether or not a redox reaction has occurred. **b** If the oxidation number of a species has increased as a result of a chemical reaction, has this species been oxidised or reduced? **c** In a reaction, the oxidation number of sulfur decreases from +6 in H_2SO_4 to –2 in H_2S. Is the sulfuric acid acting as an oxidant or a reductant?	
3	'In a redox reaction the oxidant undergoes reduction.' Explain the logic of this statement, making sure that you explain clearly the meaning of the terms 'oxidant' and 'reduction'.	

No.	Question	Answer
4	Which one or more of the following pairs of species can be accurately described as ‘conjugate redox pairs’? **A** Cu^{2+}/Cu **B** I_2/Br_2 **C** H_2SO_4/HSO_4^- **D** Sn^{4+}/Sn^{2+}	
5	**a** What is the role of an electrode in an electrolysis cell? **b** Why are some electrodes described as being ‘inert’, and under what circumstances would an inert electrode be used in an electrolysis cell? **c** Give two examples of common inert electrode materials.	
6	Is the negative electrode of an electrolysis cell the anode or the cathode?	
7	Does the electrode connected to the positive terminal of the external power source become the anode or cathode in the electrolysis cell?	
8	Explain the relationship between the cathode and anode and the cations and anions in the electrolyte of an electrolytic cell.	

Worksheet 15.2 Redox chemistry definitions	

No.	Question	Answer
9	An electrolytic cell is constructed of two inert electrodes and an electrolyte of a molten ionic compound composed of A^{2+} and B^{2-} ions. **a** Sketch a diagram of this cell, including its external power source. **b** Label on the diagram the polarity of the electrodes and where the oxidation and reduction reactions will take place. **c** Indicate the direction of the electron and ion flows in the cell.	

Worksheet 15.3 Recovering silver from solution	

NAME: **CLASS:**

INTRODUCTION

Given the high price of silver, it is desirable to have an efficient method for the recovery of 'waste' silver from aqueous solutions containing Ag^+ ions. This recovery can be achieved by using a redox reaction in which Ag^+(aq) is reduced to Ag(s). This worksheet concerns the analysis and extraction of silver from a 0.1 mol L^{-1} solution, and involves both gravimetric and volumetric techniques.

PART 1: VOLUMETRIC DETERMINATION OF SILVER ION CONCENTRATION

A 20.00 mL aliquot of 0.100 mol L^{-1} NaCl solution was pipetted into a 250 mL conical flask. Approximately 1 mL of 0.1 mol L^{-1} K_2CrO_4 solution was also added to the flask. The flask contents were then titrated with a solution of approximately 0.1 mol L^{-1} $AgNO_3$ until the first permanent red-brown colour appeared. This red-brown colour indicated that the reaction between Ag^+(aq) and Cl^-(aq) was complete (the CrO_4^{2-} ion reacts with the Ag^+ ion to form a red-brown precipitate. This reaction occurs only when there is no Cl^- ion present). The titration was completed to obtain concordant titres; an average titre of 19.32 mL was required.

No.	Question	Answer
1	What do the following words mean? aliquot pipette titrated concordant titre	
2	Write an ionic equation for the precipitation reaction between: **a** Ag^+ and Cl^- ions **b** Ag^+ and CrO_4^{2-} ions.	
3	Using the titration results, determine the concentration of the approximately 0.1 mol L^{-1} $AgNO_3$ solution.	

Worksheet 15.3 Recovering silver from solution	

No.	Question	Answer
4	Suggest two sources of error in this determination.	

PART 2: RECOVERY OF SILVER FROM SOLUTION

A 20.00 mL aliquot of the approximately 0.1 mol L^{-1} $AgNO_3$ solution was pipetted into a large test tube. A spiral of copper wire was inserted into the test tube so that most of the spiral was in the solution. The solution was stirred occasionally using the copper spiral. The test tube and spiral were allowed to stand overnight. The spiral was removed after shaking to dislodge deposited silver, and any remaining silver was washed from the coil into the test tube with distilled water. The deposited silver was collected, dried and weighed. The mass of silver recovered was 0.206 g.

No.	Question	Answer
5	How could you ensure that the recovered silver was dry?	
6	**a** Write an ionic equation for the reaction between Cu and Ag^+ ions. **b** Name the oxidant in this reaction.	
7	Determine the percentage, by mass of silver recovered from the solution.	
8	Suggest two reasons to account for the less than 100% recovery of silver from the solution.	
9	Most methods for the recovery of silver from solution involve the reduction of $Ag^+(aq)$ to $Ag(s)$. Another reductant used is the dithionite ion, $S_2O_4^{2-}$, which is oxidised to the sulfite ion, SO_3^{2-}. Write a half-equation for the oxidation of the dithionite ion (H_2O is a reactant and H^+ is a product in this reaction).	

Worksheet 15.3 Recovering silver from solution	

No.	Question	Answer
10	In an alternative process, an electric current is forced through a solution containing silver ions. Silver deposits at the negative electrode. At the positive electrode, water is oxidised to oxygen gas and hydrogen ions. Write a half-equation for this oxidation.	

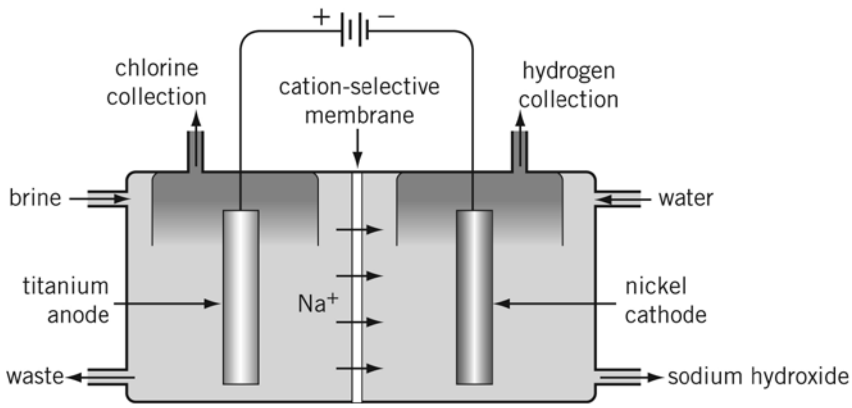
Worksheet 15.4 The chloro-alkali industry	

NAME: **CLASS:**

INTRODUCTION

The chloro-alkali industry produces two of the most important chemicals in the world: chlorine (Cl_2) and caustic soda, sodium hydroxide (NaOH). Chlorine is used mainly for the production of plastics such as PVC and other organic chemicals. It is also used for the disinfection of water supplies and in the paper industry for bleaching. Sodium hydroxide is used to make many organic and inorganic compounds. It is also used in a number of industries, including the paper industry.

Chlorine and sodium hydroxide are produced from the electrolysis of brine, concentrated sodium chloride solution. Three different types of cells, the diaphragm cell, the mercury cell and the membrane cell, have been used for this production. The membrane cell is now the preferred cell because the diaphragm cell uses asbestos and the mercury cell releases mercury into the environment. Comparatively, the membrane cell is a clean process, although the electricity it consumes often comes from fossil fuels and so the process contributes to greenhouse gas emissions. A simplified diagram of a membrane cell is shown below.



No.	Question	Answer
1	**a** Write the half-equation for the reaction that must be occurring at the negative electrode (cathode). **b** Write the half-equation for the reaction that must be occurring at the positive electrode (anode). **c** Write the overall equation for the membrane cell reaction.	

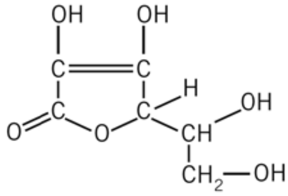
2	If 1.0 tonne of NaCl is consumed, what volume of chlorine, measured at 101.3 kPa and 0ºC, could be produced?	
3	**a** Water is a stronger reductant than chloride ions. Therefore, what are the expected products at the anode? **b** Propose a reason why the predicted substances are not produced.	
4	Sodium metal is not produced in this cell. Explain why.	
5	The polymer membrane is constructed so that only cations can pass through. Explain the advantages of this design.	
6	The left-hand anode compartment is filled continuously with brine, while the right-hand cathode compartment is filled continuously with water. Why are different liquids required?	
7	The concentration of the sodium hydroxide produced is about 30% m/m. It needs to be at least 50% m/m to be sold. **a** How could this be achieved? **b** How might this affect the cost of the products?	
8	Hydrogen is another product of the membrane cell. State one use of hydrogen gas.	

Worksheet 16.1 Carbon compounds in the home	

NAME: **CLASS:**

INTRODUCTION

Many of the chemicals you learn about at school are substances that society uses. The questions below relate to typical carbon compounds found in many homes. Look in your laundry, garage and bathroom for the products mentioned below. Examine the labels on these products to answer the questions. The functional groups present on some of the molecules below will be more complex than is required for Chemistry Unit 2B, but your teacher should be able to assist you.

No.	Question	Answer
1	**Unleaded petrol** (for lawnmowers or chain saws). This is a mixture of alkanes, including 3-ethylhexane. Give the structural formula of this compound.	
2	**Aerosol cans**. These are no longer permitted to use fluorocarbons as propellants. List examples of some aerosol products, and the propellant present (if given on the label).	
3	**Essences for cooking**, such as lemon, vanilla and peppermint. What ingredient is common to each of these? The same ingredient is common to many cough mixtures. Why is the sale of these products occasionally under scrutiny?	
4	**Vitamin tablets**. Which ingredients in these tablets sound organic? The structure of one ingredient, vitamin C, also called ascorbic acid, is shown below. Write its empirical formula. 	

5	**Sherbert**, such as in Wizz Fizz, contains citric acid. This acid is also found in citrus fruits. Draw the structural formula of a citric acid molecule.	
6	**Fingernail polish remover**. What is the active ingredient? Draw its structural formula. (Another name for it is dimethyl ketone)	
7	**Salad dressing**, vinegar-based products and Worcestershire sauce all contain acetic acid. Sketch its structural formula.	
8	**Shampoo**. List some of the ingredients from your shampoo that 'sound' organic. Can you find any of the same ingredients in other household products (such as detergents and toothpaste)?	
9	**Cooking oil and fats**. Can you find the names of the fatty acids contained in these fats and oils? Stearic acid, which occurs widely in fats, has a formula $C_{17}H_{35}COOH$. Draw a structural formula of it. Linolenic acid, which occurs widely in vegetable oils, is a polyunsaturated fatty acid. What does this mean?	
10	**Rubber** can be made from 1,3-butadiene. Give the structural formula of this molecule.	

Worksheet 16.2 Modelling and naming carbon compounds	

NAME: **CLASS:**

INTRODUCTION

The shape of organic molecules is often very significant. Many students find the two-dimensional textbook diagrams difficult to visualise, so the building of molecular models is an important exercise. There are a variety of kits available for building models.
A guiding principle in determining the shape of organic molecules is the valence shell electron pair repulsion (VSEPR) theory that electron pairs are as far away from each other as possible.

No.	Question	Answer
1	The simplest hydrocarbon is methane (CH_4). Make and sketch a model of methane. Use a protractor to measure the H–C–H angle. All four angles should be equal. What is this angle? What is the name given to the arrangement of H atoms around the C atom in this molecule?	
2	Make and sketch a model of an ethene molecule (C_2H_4). What is the H–C–H angle this time? Ethene is a planar molecule. What does this mean?	
3	Build a molecule of ethane (C_2H_6). Sketch this molecule. What is the name of the arrangement of atoms around each carbon atom?	
4	**a** What is the molecular formula of butane? **b** How many carbon-to-carbon bonds does butane have? **c** How many carbon-to-carbon bonds would octane have? **d** How many carbon-to-carbon bonds would an alkane with *n* carbon atoms have?	

5	Measure the length of one carbon-to-carbon bond in your ethane model. What distance would you expect between the first and last carbon of propane?	
6	Make a model of a propane molecule and measure this distance.	
7	Can you write a general formula to predict the length of any alkane molecule? Make several longer molecules to test your formula.	
8	Construct models of the two isomers of butane. Sketch and name each of these isomers.	
9	Construct all possible isomers of pentane. Sketch and name each of these isomers.	
10	Construct all possible isomers of hexane. Sketch and name each of these isomers.	

11	Is there a general formula for the number of isomers of an alkane with *n* carbon atoms?	
12	Sketch and name each of the molecules with the semi-structural formulas shown below. You might like to make models of some or all of these molecules. **a** $CH_3CH(CH_3)CH(CH_3)_2$ **b** $CH_2C(CH_3)_2$ **c** $CH_3CH_2C(CH_3)CHCH_3$ **d** $C(CH_3)_4$	

Worksheet 16.3 Production of ethene	

NAME: **CLASS:**

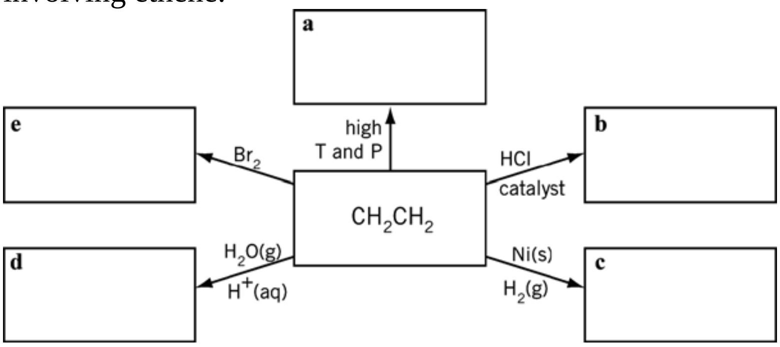
INTRODUCTION

Ethene is the leading synthetic organic chemical in terms of scale of production. Its ease of production, chemical reactivity and versatility make it well suited as the starting molecule for further syntheses. Most ethene is used to make polymers, while some is used to make ethanol, fibres and antifreeze. Since ethene does not occur naturally to any significant extent, it must be produced from hydrocarbons that are derived from natural gas and petroleum.

No.	Question	Answer
1	Ethene is insoluble in water, and has a low boiling point. Explain these properties in terms of its bonding and structure.	
2	Ethene may be produced using petroleum as the raw material. Briefly describe the process of fractional distillation of petroleum that must occur before cracking reactions to produce ethene can be carried out.	
3	Ethene can be produced by 'cracking' reactions. **a** Explain what cracking reactions are. **b** What two techniques can be used to crack molecules?	
4	**a** Write an equation for the cracking of butane to form ethene. **b** Write an equation for the cracking of butane to form molecules other than ethene.	

Worksheet 16.3

Production of ethene

No.	Question	Answer
5	Theoretically, what conditions would increase the rate of the cracking process?	
6	List the conditions actually used to produce ethene.	
7	A side reaction that interferes with the production of ethene is the production of ethyne from ethane. Write an equation for this reaction.	
8	Ethene is involved in many addition reactions. With the aid of an equation, explain what an addition reaction is.	
9	Complete the diagram by showing the products obtained in each of the addition reactions involving ethene. 	
10	Draw a flowchart to summarise the production of ethene using petroleum as a raw material.	